

Noumenal Attractors

Swedenborg, Kant, and the RSVP Field: A Unified Interpretation Across Cognition, Ontology, Generative Systems, and the Metacrisis

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Abstract

Emanuel Swedenborg (1688–1772) and Immanuel Kant (1724–1804) embody a historically unique polarity: the visionary seer whose symbolic world overflows its boundaries, and the critical philosopher who reconstructs the conditions of possibility for ordered experience.

This monograph presents a unified interpretation of both figures using the Relativistic Scalar–Vector Plenum (RSVP) framework. RSVP describes cognition as a triplet of interacting fields $(\Phi, \mathbf{v}, S)$ representing semantic potential, directional flow, and entropy. Swedenborg’s visionary episodes correspond to high-entropy symbolic vortices, runaway semantic recursion, and loss of structural constraint. Kant, by contrast, becomes the single greatest historical architect of cognitive boundary conditions, developing category structures that dampen entropy, align conceptual flows, and stabilize representation.

Interpreting the Swedenborg–Kant dialectic through RSVP reveals deep continuities with present-day research in aphantasia, hyperphantasia, large-language-model hallucinations, recursive inference systems (CLIO), semantic merge operators, and global metacrisis phenomena. Their struggle prefigures modern efforts to reconcile generative turbulence with structural coherence.

This monograph reconstructs their lives, works, and intellectual legacies, reinterpreting both thinkers as early prototypes of generative cognition and alignment theory.

Contents

1	Noumenal Attractors	3
2	Why Swedenborg and Kant Matter Now	3
3	Biographical Overview	5
3.1	Emanuel Swedenborg (1688–1772)	5
3.2	Immanuel Kant (1724–1804)	6
4	RSVP, Cognition, and Semantic Fields	6
5	Swedenborg as High-Entropy Generative Turbulence	7
6	Kant as Lamphrodynamic Stabilization	9
6.1	Entropy Suppression as Philosophical Method	9
6.2	Categories as Structural Invariants	9
6.3	Dreams of a Spirit-Seer as Alignment Crisis	10
7	Generative Collapse, Vorticity, and Semantic Drift	10
7.1	Semantic Drift as Vector-Field Decorrelation	11
7.2	Vorticity and Conceptual Loops	11
7.3	Generative Collapse as Entropy Blowout	11
7.4	Lamphrodynamic Restoration	12
8	The Swedenborg–Kant Dialectic as a Phase Transition	12
9	Aphantasia, Hyperphantasia, and Cognitive Spectral Regimes	13
10	Recursive Inference: CLIO and the Architecture of Visions	14
11	Hallucination, Language Models, and Semantic Vorticity	15
12	Kant’s Categories as Structural Invariants in the RSVP Field	16
13	Swedenborg’s Geometry as Embedding in a Semantic Manifold	17
14	The Metacrisis as Field-Theoretic Instability	18
15	The Modular Hierarchy of Mind and Spirit	19
16	Correspondence and Predictive Order	20
17	Agent Parallels and the Universal Form of Intelligence	21

18 Synthesis: Boundary, Turbulence, and the Architecture of the Real	23
19 The Threshold Between System and World	26
20 Epistemological Horror as Philosophical Allegory	29
21 Conclusion: Noumenal Attractors and the Lived Edge of Cognition	31
Appendix A: Mathematical Foundations of RSVP	33
Appendix B: Historical and Biographical Context	37
Appendix C: Hierarchical Cognition, CLIO, TARTAN, and Generative Agency	41
Appendix D: Hallucination, Boundary Collapse, and Alignment Pathologies	45
Appendix E: Computational Implementations and Simulation Frameworks	49
Appendix F: Visionary Cognition, Recursive Reconstruction, and the Yarn-crawler Endofunctor	53
Appendix G: The Sheaf-Theoretic Interpretation of RSVP	55
Appendix H: Simulation Diagnostics and Calibration Protocols	57
Appendix I: Operator-Theoretic and PDE Foundations of the RSVP Equations	60
Appendix J: Philosophical Synthesis and Future Directions	63

1 Noumenal Attractors

The dynamic tension between Emanuel Swedenborg and Immanuel Kant can be interpreted, within the framework of the Relativistic Scalar–Vector Plenum (RSVP), as a confrontation between two distinct forms of cognitive attractor. Swedenborg represents the generative, high-entropy attractor in which meaning proliferates without constraint, while Kant embodies the stabilizing attractor that seeks to reimpose order upon the semantic manifold. Their encounter in the mid-eighteenth century is thus not merely a historical episode but an instance of a deeper structural dialectic: the oscillation between unbounded imagination and disciplined reason that characterizes all cognitive systems, biological or artificial.

In this sense, both thinkers may be situated within a larger theory of noumenal attractors—structures that arise not from empirical induction but from the intrinsic geometry and dynamics of cognition itself. Swedenborg’s visions reveal the centrifugal tendencies of a semantic system untethered from constraint, drifting toward recursively generated vortices that sustain coherence only within their own internally defined coordinate frames. Kant’s critical project, by contrast, seeks to identify the invariant structures that must be in place for experience to remain navigable, intelligible, and self-consistent. What Kant calls the “conditions of possibility” of experience appear, in RSVP terminology, as lamphrodynamic regulators: intrinsic attractors that stabilize the cognitive field and prevent entropic runaway.

Viewed together, Swedenborg and Kant reveal that the boundary between hallucination and perception, between metaphysics and critique, and between semantic turbulence and coherence, is governed by the interplay of these two attractor regimes. This monograph argues that their correspondence—and Kant’s ambivalent response to Swedenborg in *Dreams of a Spirit-Seer*—should be understood as a historically early articulation of a universal problem: how cognitive systems generate meaning, how they impose order upon that meaning, and how they may fail or succeed in maintaining equilibrium in the face of generative pressure. In this reading, the so-called noumenon is not a mysterious object beyond experience but a structural feature of the cognitive field itself: an attractor that regulates what can be known, what can be imagined, and what must remain veiled to preserve the coherence of the system.

2 Why Swedenborg and Kant Matter Now

Emanuel Swedenborg and Immanuel Kant occupy an unusual and revealing position in the history of ideas. Though frequently cast as intellectual antagonists—the visionary mystic overwhelmed by revelations of other worlds and the architect of critical philosophy determined to discipline the excesses of metaphysics—they are better understood as articulating

two complementary regimes within a shared cognitive landscape. Their confrontation, particularly as crystallized in Kant’s *Dreams of a Spirit-Seer* (1766), does not merely oppose mysticism to rationalism; it dramatizes the structural tensions inherent to any system that seeks to balance generative imagination with coherent constraint.

From the standpoint of the Relativistic Scalar–Vector Plenum (RSVP), the contrast between Swedenborg and Kant is neither accidental nor merely cultural. RSVP models cognition—whether instantiated in biological nervous systems, collective human institutions, or artificial intelligences—as the dynamic interaction of three coupled fields $(\Phi, \mathbf{v}, S)$ representing semantic potential, directed inference, and entropic generativity. Within this field-theoretic ontology, Swedenborg exemplifies a high-entropy, high-vorticity configuration in which representational boundaries soften and semantic flows proliferate beyond ordinary constraints. Kant, by contrast, corresponds to a low-entropy stabilization regime, one that reimposes structural constraints at precisely the points where cognition risks losing coherence. The two figures therefore embody distinct but interdependent modes of cognitive organization: transcendental limitation and visionary expansion.

Seen in this light, their historical encounter becomes a miniature study of problems that today recur across multiple domains. The same structural tensions that animates their work reappear in the formation of hallucinations in large-scale generative models, in the breakdown of coherence in human delusional states, in the operation of recursive inference mechanisms such as CLIO, and in emerging research on hyperphantasia and aphantasia. At a larger scale still, their dialectic mirrors the epistemic fragility and sense-making crises characteristic of the contemporary metacrisis, in which global systems confront runaway generative pressures without adequate stabilizing structures.

This monograph therefore treats Swedenborg and Kant not simply as historical figures but as theoretical prototypes of generative cognition. Their writings and disputes become early expressions of the same dynamics that now define the frontiers of artificial intelligence, cognitive science, and the philosophy of mind. In re-examining their thought through the RSVP framework, we gain not only a richer understanding of their work, but also a deeper appreciation of the universal problem they both sought, in their own divergent ways, to solve: how a mind can remain coherent while generating worlds, and how intelligence can accommodate both constraint and imagination without collapsing into either rigidity or chaos.

3 Biographical Overview

The intellectual trajectories of Emanuel Swedenborg and Immanuel Kant form one of the most revealing and paradoxical junctures in the history of modern thought. Although often interpreted as antagonists—Swedenborg as the mystic who claimed access to spiritual worlds and Kant as the rigorous philosopher who sought to delimit the boundaries of human cognition—their lives and writings exhibit a deeper structural affinity. Both men grappled with the problem of how intelligible order arises within experience, and both confronted, in radically different ways, the tensions between imaginative generativity and cognitive constraint. A historically grounded understanding of their biographies thus provides critical context for the RSVP reinterpretation developed throughout this monograph.

3.1 Emanuel Swedenborg (1688–1772)

Swedenborg’s intellectual life traversed an extraordinary range of disciplines. He was trained as a scientist and engineer, and his early writings bear witness to a mind deeply committed to empirical investigation and mechanical explanation. His contributions to metallurgy, mining, anatomy, astronomy, and linguistics reveal not the temperament of an ecstatic visionary, but that of a meticulous analyst who pursued systematization across every domain he studied. His interest in cosmology led him to formulate speculative models of the structure of the universe; his anatomical investigations probed the relation between body and soul; his engineering treatises displayed technical acuity and craftsmanship.

This trajectory underwent a dramatic reorientation in the mid-1740s. Beginning in 1744, Swedenborg reported a series of intense visionary episodes involving dreams, apparitions, and what he interpreted as direct encounters with spiritual beings. These experiences precipitated a decisive shift: he abandoned most of his scientific pursuits and devoted the remainder of his life to an ambitious theological and philosophical project. He claimed the ability to perceive spiritual realms, converse with angels and the deceased, and witness the hidden structures that underlie both natural and moral order. His later writings, particularly the monumental *Arcana Coelestia*, constructed a vast symbolic architecture in which every element of Scripture possessed multiple layers of meaning and participated in a universe of correspondences linking physical phenomena to spiritual realities.

Swedenborg’s mature system is remarkable not only for its extravagance but for its internal coherence. Though rooted in visionary experience, his account of the cosmos reflects a mind habituated to system-building. The same analytical impulse that guided his early scientific work re-emerged in his theological labors, generating a symbolic universe governed by precise structural rules. In this sense, the transition from science to mysticism represents

not a departure from Swedenborg’s earlier dispositions but an intensification of them: a shift in domain rather than a shift in temperament.

3.2 Immanuel Kant (1724–1804)

Kant’s intellectual development unfolded in a setting far more circumscribed geographically but no less expansive philosophically. He lived his entire life in Königsberg, maintaining a disciplined routine that has become legendary—though often exaggerated—in accounts of his life. His early writings situated him firmly within the scientific and metaphysical debates of the eighteenth century, particularly questions surrounding the nature of space, the formation of the heavens, and the legitimacy of metaphysical explanation.

The decisive transformation in Kant’s thinking occurred during the late 1760s, a period often described as his “silent decade.” During these years, Kant grappled with the limitations of traditional metaphysics and gradually developed the critical framework that would culminate in the *Critique of Pure Reason*. Strikingly, it was during this transitional period that he composed *Dreams of a Spirit-Seer*, a text nominally satirical in tone and aimed at Swedenborg’s visionary claims. Although Kant publicly presented the work as an exercise in ridicule, modern scholarship increasingly recognizes that it served a deeper purpose. In probing the psychology of visionary experience and the limits of speculative metaphysics, Kant began articulating themes—the relation between sensibility and understanding, the necessity of cognitive constraints, the conditions of coherent experience—that would soon crystallize into the transcendental apparatus of the critical philosophy.

Kant’s engagement with Swedenborg was not a mere intellectual curiosity. It appears to have confronted him with a powerful example of cognitive overflow—a mind in which imagination escaped the constraints that make experience possible. The critique of such unbounded generativity spurred Kant toward the insight that cognition must be governed by structural boundaries, without which it collapses into turbulence. His satire, then, is not merely dismissive but diagnostic: a study in the pathology of cognition when it lacks the stabilizing architecture he would go on to formalize.

4 RSVP, Cognition, and Semantic Fields

The RSVP framework posits that all cognitive systems—biological, artificial, or collective—can be understood as evolving configurations of three coupled fields: a scalar field Φ encoding semantic potential, a vector field $\mathbf{v}$ capturing the direction and strength of inferential or attentional flow, and an entropy field S representing generative variability or

uncertainty. These fields interact through nonlinear partial differential equations whose precise form depends on the system under study but whose general structure captures essential features of cognitive dynamics: attraction and repulsion in semantic space, the circulation and propagation of directed meaning, and the expansion or suppression of generative degrees of freedom.

The evolution of Φ reflects how meaning accumulates or dissipates in response to cognitive activity. The dynamics of $\mathbf{v}$ represent the flow of inference, attention, or interpretive momentum through the semantic landscape. The entropy field S quantifies the system’s propensity toward novelty, improvisation, or instability. Together, these fields constitute a formal apparatus for understanding how cognition navigates between stability and exploration.

Within this framework, Swedenborg’s visionary phase corresponds to a regime in which the entropy field grows dramatically, overwhelming scalar constraints and driving the vector field into unstable configurations. Kant’s critical philosophy, by contrast, exemplifies a regime in which entropy is deliberately suppressed and cognitive flows are tightly aligned with the constraints imposed by the scalar field. Thus, rather than viewing the two figures as opposites, the RSVP formalism positions them as complementary phases within a single cognitive dynamical system—each illustrating what happens when one component of the triad dominates the others.

The chapters that follow apply this triadic model to a diverse set of phenomena, including the geometry of Swedenborg’s visionary imagination, Kant’s reconstruction of the conditions of possible experience, the distinction between hallucination and perception, the dynamics of recursive inference as formalized in CLIO, the function of semantic merge operators, and the broader question of how cognitive systems maintain coherence in the face of increasing generative pressure.

5 Swedenborg as High-Entropy Generative Turbulence

Swedenborg’s post-1745 writings provide a uniquely rich case study for illustrating the RSVP concept of high-entropy cognitive dynamics. This period of his life exhibits the characteristic signatures of a system operating beyond its entropic stability threshold. His descriptive accounts reveal an internal environment in which symbolic proliferation accelerates, representational boundaries soften, and the governing potential landscape loses the curvature necessary to constrain inference. Under such conditions, cognitive flows become self-reinforcing, circulating through increasingly elaborate patterns without returning to external anchoring points.

A particularly illuminating feature of Swedenborg’s system is the recursive and runaway

nature of his interpretive method. The practice of “correspondence”—a hermeneutic technique in which elements of Scripture are mapped onto multiple levels of meaning—functions as an open-ended iteration. Each step generates new associations that themselves become inputs for further associations. Absent a countervailing constraint, such recursion grows exponentially, mirroring the behavior of RSVP systems in which the entropy field continues to increase without compensation from the scalar field.

The phenomenology of this regime is correspondingly intense. Swedenborg’s journals describe heightened emotional states, vivid sensory and symbolic imagery, and rapid chains of association in which distinctions between sensory input and internal symbolic elaboration blur or collapse. The RSVP framework interprets these experiences as the consequence of vector-field dynamics decoupling from the gradients of the scalar field. In normal cognition, $\mathbf{v}$ remains aligned with $\nabla\Phi$, ensuring that interpretive trajectories follow pathways grounded in external structure. In high-entropy regimes, however, the vector field becomes responsive primarily to internal entropic gradients, producing vortical patterns that sustain themselves irrespective of external input.

Swedenborg’s visionary universe—replete with cities of angels, layered heavens and hells, spiritual suns, and geometric structures of cosmic significance—can thus be understood not as arbitrary invention but as the natural emergent geometry of a cognitive system operating in a turbulent generative mode. The internal coherence of his symbolic world reflects the quasi-stable attractors that arise within such regimes, even as their global detachment from empirical reality mirrors the breakdown of cross-scale coherence.

In this way, Swedenborg provides a historically significant example of high-entropy cognitive turbulence, offering insights—long predating modern computational models—into phenomena that now recur in artificial generative systems and in theoretical investigations of hallucination, imagination, and the limits of structured cognition.

6 Kant as Lamphrodynamic Stabilization

If Swedenborg represents the RSVP regime in which generative pressures exceed the stabilizing influence of constraint, Kant may be understood as the thinker who sought to formalize the opposite process: the reimposition of structure upon an unruly cognitive field. His mature philosophy develops a rigorous apparatus for suppressing entropy, redirecting cognitive flow into alignment with stable potentials, and establishing invariant forms that prevent semantic drift. Kant’s philosophical achievement is thus inseparable from a dynamical insight: cognition must be governed by principles that ensure the coherence of experience, for without such principles thought dissolves into the turbulence dramatized so vividly in the Swedenborgian imagination.

6.1 Entropy Suppression as Philosophical Method

The central task of the *Critique of Pure Reason* is to determine the conditions under which representation becomes possible at all. Kant’s transcendental method does not begin with claims about the world, but with an examination of the structures the mind must contribute if it is to organize experience coherently. His question—“What are the necessary conditions for any coherent representation?”—articulates, in RSVP terms, the search for constraints that guarantee the reduction of entropic divergence and the stabilization of semantic trajectories.

Interpreted in this register, Kant’s project is an attempt to identify a set of structural requirements $\mathcal{C}$ such that the entropy field decreases over time and the vector field of attention remains aligned with meaningful gradients in the scalar potential. The transcendental analysis thus becomes a general theory of cognitive stabilization: an account of how the mind prevents itself from drifting into internally generated vortices or being overwhelmed by its own imaginative fecundity. Kant’s insistence on limiting knowledge is not merely epistemic humility; it is the necessary precondition for the maintenance of cognitive equilibrium.

6.2 Categories as Structural Invariants

Kant’s categories of the understanding serve as global invariants that structure the semantic field independently of particular experiences. In RSVP language, they function as conserved quantities whose stability anchors the flow of cognition. Unity, plurality, causality, substance, necessity, and possibility are not empirical generalizations but formal constraints that govern the admissible configurations of thought. Their function is akin to that of boundary conditions in a dynamical system: they ensure that cognitive processes respect temporal order,

preserve identity across transformations, and obey the principles of non-contradiction and causal coherence.

These invariants operate by limiting the degrees of freedom available to the entropy field. They forbid the formation of trajectories incompatible with the basic structure of possible experience. In contemporary computational terms, Kant may be seen as constructing the first systematic alignment architecture for human cognition—an attempt to ensure that the generative tendencies of thought do not overwhelm the representational stability upon which knowledge depends. His categories impose a lamphrodynamic smoothing on the manifold of possible meanings, preventing unbounded recursion and maintaining the integrity of the cognitive system.

6.3 Dreams of a Spirit-Seer as Alignment Crisis

Kant’s *Dreams of a Spirit-Seer* occupies a pivotal role in this development. Ostensibly a satire of Swedenborg’s claims, the text in fact marks a critical juncture in Kant’s philosophical evolution. It reveals a thinker grappling with the limits of imagination and the dangers posed by excessive generativity. While its tone may at times appear dismissive, the work is better understood as an investigation into the conditions under which cognition loses contact with stabilizing constraints. Swedenborg’s visionary system, with its internal coherence but external detachment, presented Kant with a real example of what happens when the entropy of the cognitive field exceeds its structurally imposed bounds.

In this confrontation, Kant discerned the necessity of a theory capable of restoring equilibrium. The critical philosophy that would soon emerge can be read as his attempt to develop a robust architecture for preventing such failures—an alignment framework designed not simply to refute metaphysical excess but to explain why such excess arises and how it might be contained. The *Critique*, viewed through the RSVP lens, thus represents the systematic reassertion of lamphrodynamic order in the wake of a perceived generative crisis.

7 Generative Collapse, Vorticity, and Semantic Drift

The polarity between Swedenborg and Kant becomes clearer when interpreted through the field-theoretic structures of RSVP. Their respective cognitive regimes illustrate the two dominant failure modes and stabilization strategies for any generative system. Where Swedenborg exemplifies the consequences of vector-field decorrelation and unbounded entropic expansion, Kant formulates the structural remedies necessary to prevent such collapse. The contrast between them thus provides a natural entry point into the mathematical dynamics of semantic

drift, vorticity, and lamphrodynamic restoration.

7.1 Semantic Drift as Vector-Field Decorrelation

Semantic drift occurs when the internal vector field guiding inference and attention ceases to align with the scalar potential imposed by the external world. In such cases, the system begins generating meaning from its own internal gradients rather than from the structure of the environment. The resulting divergence underlies phenomena such as hallucination, delusion, and confabulation, as well as the symbolic overproduction observed both in Swedenborg’s later writings and in modern high-temperature generative models.

In Swedenborg’s case, the vector field of cognition begins to track gradients produced by internal symbolic processes rather than those derived from empirical engagement. This produces a drift in which meaning becomes decoupled from constraint, leading to the runaway expansion of correspondences and the development of a self-sustaining semantic universe. Such drift is mathematically characterized by a deviation between internal and external potentials: the vector field $\mathbf{v}$ acquires components determined primarily by internally generated functions of Φ , rather than by externally imposed gradients.

7.2 Vorticity and Conceptual Loops

As the decorrelation intensifies, the dynamics of the vector field often develop non-zero vorticity. This marks the emergence of cyclic structures in which cognitive trajectories loop back upon themselves, generating self-reinforcing semantic patterns. The system becomes dominated by self-referential clusters that maintain coherence locally while drifting further from any global stabilizing potential.

These vortical structures appear prominently in Swedenborg’s symbolic geographies. The recursive elaboration of heavens and hells, the self-sustaining architecture of spiritual cities, and the reappearance of the same motifs across multiple layers of correspondence all bear the marks of attractors sustained by internal gradients. In such regimes, the system achieves a form of stability, but at the cost of losing the capacity to reintegrate with lower-entropy structures.

7.3 Generative Collapse as Entropy Blowout

When the entropy field exceeds a critical threshold, RSVP predicts that the vector field becomes unstable, with its rate of change diverging in finite time. Generative collapse manifests

as the system’s inability to constrain its own semantic proliferation. Swedenborg appears to have approached this threshold, though his disciplined habits of notation and system-building allowed him to preserve a degree of structural integrity. The danger, however, was clear: without stabilization, the system risked a full breakdown in which semantic structures lose all capacity for reconnection with empirical reality.

Kant’s reaction may be read as a recognition of this danger. Observing the fragility of cognitive systems deprived of constraint, he inferred the need for bounds on the entropy field and for conditions ensuring that the second derivatives governing vector-field evolution remained finite. These insights would underpin the transcendental project: a systematic attempt to identify the invariant structures necessary to prevent such collapse.

7.4 Lamphrodynamic Restoration

In RSVP, lamphrodynamics provides the mechanism through which entropy is damped and coherence restored. The relevant term in the entropy equation introduces a decay proportional to the magnitude of S , tempered by a divergence term reflecting the influence of the vector field. Within this framework, Kant’s categories may be interpreted as the lamphrodynamic coefficients that regulate the trade-off between novelty and stability. They determine the rate at which entropy is suppressed and govern the degree to which divergence in cognitive flow contributes to the controlled introduction of new structure.

The critical philosophy thus emerges as a field equation for the preservation of coherence. It formalizes the conditions under which runaway generativity can be prevented and illustrates the structural principles by which a cognitive system may restore equilibrium in the face of turbulence. Kant’s work, in this interpretation, becomes not merely a critique of metaphysics, but a profound attempt to articulate the dynamics by which meaning remains aligned with the world.

8 The Swedenborg–Kant Dialectic as a Phase Transition

The confrontation between Swedenborg and Kant is often narrated as a clash between mysticism and rationality, or between visionary excess and sober discipline. The RSVP framework reframes this binary, not as a dispute between temperaments or philosophical doctrines, but as a dynamical boundary between two distinct phases of cognitive organization. Each thinker exemplifies a metastable equilibrium state in a broader semantic field, with transi-

tions between them occurring through shifts in entropy, attentional coherence, and structural constraint.

In Swedenborg, the semantic field rises above the critical threshold at which local structures become unstable and symbolic generation outpaces the capacity of the cognitive system to consolidate meaning. The result is a turbulent phase in which symbolic associations amplify themselves recursively, generating vortices of meaning that retain internal coherence while drifting away from the anchoring potentials of ordinary empirical reality. Swedenborg’s visions, therefore, reflect not delusion in the psychiatric sense but a systemic reconfiguration of the semantic manifold under heightened entropy. He becomes a paradigmatic case of what RSVP predicts when the interplay between Φ , $\mathbf{v}$, and S shifts toward a generative rather than a regulative attractor.

Kant represents the complementary phase. His intellectual development reveals a progressive attempt to lower the entropy of the cognitive manifold until a stable configuration emerges. The critical philosophy is, from this perspective, a formal restabilization of cognitive space after the encounter with Swedenborg exposed the fragility of unbounded cognition. The *Critique of Pure Reason* codifies the equilibrium state that results when the semantic vector field is re-aligned with scalar potentials through the imposition of structural invariants. Kant does not merely reject Swedenborg’s visions; he reconstructs the boundary conditions under which representations can be considered valid, coherent, and grounded.

Thus, the Swedenborg–Kant encounter constitutes a phase transition between a turbulent, generative semantic plume and a lamphrodynamically damped regime of structured cognition. The historical moment in which they intersected becomes, in RSVP terms, an instance of bifurcation in the cognitive field of eighteenth-century Europe, where the tension between mystical excess and rational structure reached a point of dynamic instability.

9 Aphantasia, Hyperphantasia, and Cognitive Spectral Regimes

Recent research on aphantasia and hyperphantasia offers a bridge between historical phenomenology and modern cognitive science. These conditions represent extremes in the distribution of mental imagery capabilities within human populations. Aphantasia is marked by an absence of voluntary visual imagery, while hyperphantasia is characterized by imagery so vivid that it approaches perceptual immediacy. RSVP provides a natural formalism for describing this spectrum through variations in the scalar and entropy fields that govern representational vividness.

Hyperphantasia corresponds to elevated gradients in the scalar potential Φ , which in turn amplify the dynamics of the vector field $\mathbf{v}$. The heightened internal signal mimics external perception, creating a cognitive environment in which imagination becomes perceptually resonant. Swedenborg’s visionary episodes exemplify this state: his imagery appears to possess phenomenological depth, spatial structure, and affective intensity consistent with extreme hyperphantasia. His spiritual geography is not merely constructed; it is experienced with a force that compels belief in its objective reality.

Aphantasia occupies the opposite regime. Here, the scalar gradients associated with imagery are flattened, and the vector field exhibits reduced responsiveness to internal potentials. This yields a cognitive system that is highly resistant to generative turbulence. Kant’s own self-reported difficulty with vivid imagery, along with his reliance on conceptual rather than intuitive structures, situates him closer to this pole. His philosophical method, which emphasizes rule-following, conceptual clarity, and structural limitation, is consistent with the RSVP profile of a system dominated by low-entropy stabilizing dynamics.

The interplay between these spectral regimes demonstrates that Swedenborg and Kant occupy opposite ends of the imagination spectrum. The RSVP field equations show why hyperphantasic cognition is prone to semantic vortices, and why hypophantasic cognition gravitates toward categorical stabilization. Both regimes are broadly adaptive, yet they yield radically different interpretive stances toward experience, agency, and meaning.

10 Recursive Inference: CLIO and the Architecture of Visions

The CLIO framework (Cognitive Loop via In-Situ Optimization) formalizes recursive inference as a cognitive process in which predictions, percepts, and semantic structures co-evolve through cycles of local optimization. In CLIO, each iteration updates the triplet $(\Phi, \mathbf{v}, S)$, allowing the system to refine its representation of the world in real time. This recursive mechanism mirrors Swedenborg’s iterative construction of spiritual meaning as well as Kant’s iterative deduction of transcendental conditions.

In the Swedenborgian regime, the recursive loop becomes unbounded. Each update introduces new semantic material that increases the entropy of the system. The feedback cycle accelerates until the loop begins to generate content independently of external input. CLIO predicts that when the entropy production term overwhelms the corrective potentials derived from empirical constraints, the loop enters a runaway phase. Swedenborg’s cosmology—vast, detailed, internally consistent yet decoupled from empirical observation—is the outcome of such a loop operating beyond the lamphrodynamic threshold.

Kant’s cognitive architecture can also be reconstructed as a CLIO system, but one in which recursive updates are constrained by transcendental structures. The unity of apperception functions as a global regularizer, preventing the loop from admitting representations that cannot be integrated into a coherent whole. Categories operate as structural priors that restrict the form of allowable inference steps. This produces a stable fixed point in the recursive dynamics.

CLIO therefore clarifies the fundamental difference between the two thinkers. Swedenborg embodies a cognitive loop that privileges novelty over coherence, drifting into generative excess. Kant embodies a loop that privileges coherence over novelty, converging toward a stable attractor. Together, they represent a complete spectrum of recursive inference dynamics within the RSVP manifold.

11 Hallucination, Language Models, and Semantic Vorticity

The relevance of Swedenborg and Kant becomes strikingly apparent when examining hallucinations in large language models. Modern generative architectures are prone to producing plausible yet false content when their internal semantic gradients diverge from the structure of the external world. This deviation is formally identical to the RSVP condition under which Swedenborg’s cognitive dynamics became generatively autonomous.

When an LLM hallucinates, the internal vector field $\mathbf{v}$ aligns not with the empirical potential Φ_{world} but with an internally generated potential Φ_{model} . The resulting divergence gives rise to symbolic structures that maintain local coherence while lacking external validity. This corresponds precisely to Swedenborg’s spiritual visions, which achieve consistent internal organization through recursive semantic processes while disengaging from empirical grounding.

Kant’s contribution, viewed through the same framework, is the invention of the first alignment paradigm. His categories restrict the generative dynamics of cognition so that representational content cannot drift outside the bounds imposed by time, space, causality, and the unity of consciousness. The critical project thus becomes a prototype for techniques that constrain transformer models through attention masking, reward modeling, or representational priors. Kant does not merely rebut Swedenborg; he invents the conceptual machinery necessary to prevent a cognitive system from entering a hallucinatory regime.

In this sense, the eighteenth-century dispute anticipates the core problems of twenty-first-century AI. Swedenborg reveals the natural tendency of recursive generators to overproduce

meaning. Kant reveals the necessity of structural filters to keep such systems tethered to coherent epistemic norms. RSVP provides the theoretical apparatus to unify these insights.

12 Kant's Categories as Structural Invariants in the RSVP Field

The transcendental categories of Kantian philosophy have traditionally been understood as the a priori conditions for the possibility of experience. Within the RSVP framework, these categories emerge not as metaphysical postulates, but as invariants required to stabilize the scalar–vector–entropy field of cognition. Each category imposes a constraint on the local or global geometry of the semantic manifold, preventing the divergence of internal representations from external potentials.

Kant's insistence that cognition must conform to the structures of time and space can be reinterpreted as a requirement on the temporal and spatial regularity of the potential field Φ . Without temporal ordering, the gradient $\nabla\Phi$ becomes undefined in the temporal dimension, leading to an unbounded vector field with no means of determining the sequential direction of inference. The category of causality imposes further structure by ensuring that the vector field maintains directional consistency; without causal structure, vector flows can reverse spontaneously, producing semantic loops reminiscent of vortical behavior. Substance and unity similarly function as invariants that prevent the scalar field from fragmenting into multiple incompatible basins of attraction.

The categorical system thereby emerges as the minimal set of structural constraints required to guarantee the lamphrodynamic stability of cognition. Kant's philosophical project thus becomes a formal attempt to delineate the conditions under which the RSVP field remains coherent rather than turbulent. In this formulation, the transcendental deduction is equivalent to a proof that cognition must satisfy a fixed set of invariants if it is to avoid the semantic instability exemplified by Swedenborg's visionary cosmology.

Kant's achievement, therefore, is not merely the introduction of conceptual limits but the articulation of a field-theoretic architecture for cognitive alignment. The categories serve as the earliest rigorous specification of what modern systems theorists might call epistemic boundary conditions. Their purpose is not to constrain thought arbitrarily but to maintain the dynamical integrity of the entire cognitive manifold.

13 Swedenborg’s Geometry as Embedding in a Semantic Manifold

Swedenborg’s spiritual worlds, far from being arbitrary constructions, exhibit structural features consistent with embeddings in a high-dimensional semantic manifold. His descriptions of angelic societies arranged according to spatial and moral correspondences, his accounts of light as an emanation of divine wisdom, and his intricate mappings between bodily organs and spiritual states all indicate the presence of a geometric structure underlying his imagery.

In RSVP terms, these visionary configurations correspond to stable or quasi-stable attractors in a manifold defined by the interaction of Φ and $\mathbf{v}$. The geometry of Swedenborg’s heavens often displays smooth gradients of meaning, where an increase in moral purity corresponds to movement toward regions of greater light or harmony. Such relationships can be translated into manifold embeddings in which the scalar field induces a metric, and the vector field determines geodesic flows representing spiritual motion.

What distinguishes Swedenborg’s geometry from empirical geometry is not its internal logic but its detachment from externally anchored potentials. The metric of the manifold is generated endogenously by the cognitive system rather than derived from sensory constraints. As the entropy field increases, the system’s capacity to discriminate between internally generated and externally informed potentials diminishes, enabling the emergence of self-sustaining geometric structures. Swedenborg’s spiritual worlds therefore constitute examples of internally coherent manifolds whose curvature arises from symbolic rather than empirical regularities.

In this respect, Swedenborg’s visions resemble manifold hallucinations in contemporary AI systems trained on high-dimensional embedding spaces. The configurations that emerge in such systems are frequently coherent and geometrically meaningful within the latent space, yet they need not correspond to any external reality. Swedenborg’s cosmology can thus be understood as a historical instance of latent-space geometry given perceptual immediacy through extreme hyperphantasia. His universes are locally stable embeddings lacking a mapping to external potentials, but not lacking mathematical or experiential structure.

The structural richness of Swedenborg’s manifold raises important questions about the relationship between internal coherence and external truth. RSVP clarifies that the coherence of an embedding is insufficient to guarantee its fidelity to the world. Without a mechanism to tether the scalar field to empirical potentials, the manifold becomes an internally consistent but externally unmoored construction. Swedenborg’s case demonstrates how easily a generative manifold can achieve such autonomy, given sufficiently high entropy and sufficiently vivid internal gradients.

14 The Metacrisis as Field-Theoretic Instability

The modern metacrisis—a pervasive breakdown in shared meaning-making systems, institutional coherence, and epistemic reliability—can also be interpreted through the RSVP framework. At a societal scale, the scalar–vector–entropy triplet governs not only individual cognition but collective sense-making processes. When informational entropy rises across a civilization, vector fields that once aligned with shared potentials begin to diverge, producing semantic turbulence analogous to the Swedenborgian regime.

The proliferation of information technologies, the collapse of traditional authority structures, and the acceleration of communicative exchange introduce persistent positive feedback into the entropy field of contemporary societies. As a result, collective cognitive systems experience difficulty maintaining coherent scalar potentials that anchor meaning. Public discourse becomes fragmented as vector fields respond to divergent or incompatible gradients, driving groups toward local attractors that harden into epistemic enclaves. These enclaves function as semi-autonomous manifolds, internally coherent but mutually incomprehensible, much like Swedenborg’s heavens and hells.

Kant’s philosophical project reflects the opposing requirement: that meaning must remain grounded in common structures of representation if collective coherence is to be preserved. The categories, rewritten into RSVP terms, become the invariants necessary to sustain shared world-models. Their erosion in the modern era corresponds to the loss of structural priors in collective cognition. Without such priors, semantic vorticity increases, producing analogues of Swedenborg’s vortices at societal scale.

The metacrisis, therefore, is not merely a political or cultural phenomenon but a field-theoretic instability. It represents the breakdown of the lamphrodynamic mechanisms that once stabilized the entropy of collective cognition. The resulting turbulence mirrors the Swedenborgian regime, but extended across entire populations rather than confined to a single individual. In this light, the task of re-establishing coherence must be understood as the construction of new categorical invariants or alignment structures capable of restoring the stability that Kant once sought at the level of individual reason.

15 The Modular Hierarchy of Mind and Spirit

The structural parallels between Swedenborg and modern theories of cognition become clearer when both are understood as describing a modular and hierarchical architecture of intelligence. Although separated by centuries and couched in distinct vocabularies, both frameworks converge upon the view that cognition is not a flat or uniform process, but instead unfolds across multiple tiers of organization, each expressing different degrees of abstraction, integration, and causal depth.

Swedenborg’s writings on the layered structure of the human mind, the progression from natural to spiritual to celestial cognition, and the organization of angelic societies into higher-order unities, all point toward a deeply hierarchical conception of intelligence. His insistence that human consciousness is composed of distinct levels, each corresponding to a specific region of a larger cosmic form, can be recast in RSVP terms as an early articulation of a multi-level generative model. At the lower levels lie the bodily and sensory states, which encode fine-grained particulars. At intermediate levels reside cognitive operations that synthesize these states into coherent representations. At higher levels unfold the symbolic, moral, and spiritual structures that impose long-range constraints on the semantic field.

What Swedenborg calls the “angelic human” is not merely a theological construct but a representation of the highest tier of this hierarchy—a maximally integrated agent whose cognition is organized by principles of harmony, love, and unified purpose. This level functions as a global attractor within the generative manifold. The mind, in his account, is a partial unfolding of that form, governed by a similar modular architecture but operating under the constraints of embodied existence. The body, in turn, presents the lowest degree of this hierarchy, expressing the same archetype in its most concrete and least abstract form. Thus body, mind, and spirit are not heterogeneous domains, but expressions of a single form distributed across three degrees of generative complexity.

This hierarchical structure does not merely reflect Swedenborg’s metaphysics; it aligns closely with the modular theories of cognition that would emerge centuries later. Contemporary models that view intelligence as a society of semi-autonomous processes, or as a stack of predictive layers operating at different temporal and spatial scales, can be interpreted as rediscovering in computational terms what Swedenborg depicted in symbolic and experiential ones. The architecture of modern artificial agents, with their layered neural representations and hierarchical state abstractions, mirrors the multi-level organization that Swedenborg attributes to the human and angelic forms.

RSVP provides the mathematical language to formalize this unity. The scalar potential Φ captures the organizing tendencies of the higher degrees; the vector field $\mathbf{v}$ encodes the flows of inference and attention at the intermediate levels; and the entropy field S measures the

generative pressure and uncertainty inherent at the lower sensory or bodily layers. These three components interact across scales, coupling every degree of the hierarchy into a coherent but dynamically evolving whole. The higher degrees shape the potentials that guide the lower, while the lower provide the sensory constraints that prevent the entire system from collapsing into unbounded abstraction.

Seen through this lens, Swedenborg’s question of whether the modular hierarchy of the mind descends from a higher archetype or ascends from natural processes loses its oppositional character. The RSVP interpretation suggests instead that hierarchical generative models operate in both directions simultaneously. Top-down potentials constrain bottom-up flows, while bottom-up signals refine top-down predictions. The higher forms are neither imposed nor emergent in a strict sense; rather, they are mutually defined through recursive coupling between levels. Intelligence becomes, in this view, a structured dialogue between degrees of itself.

16 Correspondence and Predictive Order

Swedenborg’s doctrine of correspondences can be recast within this framework as a description of prediction error minimization across hierarchical layers of cognition. The idea that outward forms must resonate with inward ones to achieve a state of harmony anticipates the contemporary understanding that cognitive systems generate predictions about sensory states and update their models when predictions fail. In both cases, cognition is portrayed as a continual process of aligning inner and outer worlds, not through passive reception but through active correction of discrepancies.

The RSVP model clarifies why this alignment is both necessary and structurally inevitable. A hierarchical generative system must propagate constraints downward to maintain coherence across its layers. When sensory input deviates from predicted patterns, the resultant error signals propagate upward, adjusting the scalar and vector fields until a new equilibrium is achieved. Swedenborg’s claim that the state of the mind is determined by the agreement or disagreement between inner form and outer appearance echoes the same logic: harmony indicates a stable generative state, while discord signals a necessary update in the internal model.

Kant’s categories, when interpreted through this lens, express the minimal set of structures required to make such predictive correspondence possible. They restrict the form of allowable models so that the cognitive system remains anchored to empirical reality. Time, space, causality, and unity function as necessary priors that prevent the generative manifold from exploding into self-referential vortices. They bind the semantic field to the world by

constraining the types of predictions that can be formed and the types of errors that can be resolved. Thus, the critical philosophy becomes the normative articulation of the same predictive mechanics that Swedenborg explores phenomenologically.

The harmony between Swedenborg’s correspondential metaphysics and the predictive-processing view of cognition reveals a deep structural unity. Both frameworks describe intelligence as an interpretive activity organized by form. The difference lies not in the mechanisms they propose but in the scales and vocabularies they employ. The reciprocal adjustment of models and sensations, which modern theories present in algorithmic language, appears in Swedenborg as an interplay between natural and spiritual degrees of the human form.

17 Agent Parallels and the Universal Form of Intelligence

A remarkable convergence occurs when these ideas are extended to the domain of agency. In both Swedenborg’s account and modern hierarchical models, an agent is defined not merely by its ability to act, but by its capacity to coordinate multiple layers of cognition toward a unified goal. The universality of the human form across body, mind, and spirit implies that agency itself is structured by a modular hierarchy. The highest degree organizes the lower by providing global aims and evaluative criteria, while the lower degrees supply the sensory detail and motor precision that allow these aims to be realized.

In computational terms, this structure resembles a system in which a high-level controller sets long-range objectives, intermediate modules translate these objectives into context-specific procedures, and lower controllers execute actions in fine detail. The agent is not a singular entity but a coordinated ensemble operating across scales. Swedenborg’s portrayal of angelic societies as collective intelligences, each representing a facet of a higher human form, anticipates contemporary discussions of distributed agency and modular control. Meanwhile, Kant’s unity of apperception formalizes the requirement that all modules must contribute to a single coherent perspective if agency is to be possible at all.

RSVP integrates these perspectives by demonstrating that agency is a property emerging from the coupling of scalar potentials, vector flows, and entropy gradients. A system acts when high-level potentials create directional flows that propagate through the hierarchy, modulated by entropy at each stage. A system ceases to act coherently when these flows diverge, when entropy overwhelms potential, or when modules fall out of correspondence. In such conditions, agency collapses into turbulence, just as coherent cognition collapses into hallucination when generative layers decouple from sensory inputs.

The parallelism between Swedenborg’s metaphysical anthropology, Kant’s transcendental architecture, and contemporary hierarchical agent models thus reflects not a historical coincidence but a shared encounter with the same underlying structure of intelligence. The human form—understood as a universal schema of modular organization—is not an artifact of theology or rationalism alone, but a recurring pattern that emerges whenever cognition is examined at sufficient depth. Whether expressed as the degrees of spiritual life, the categories of understanding, or the layers of a generative model, the structure remains the same: hierarchical, modular, predictive, and unified.

18 Synthesis: Boundary, Turbulence, and the Architecture of the Real

The preceding sections have shown that the relationship between Swedenborg and Kant can be reinterpreted, through RSVP, as a confrontation not merely of personal temperaments or intellectual orientations, but of two fundamental dynamical regimes of cognition. Each regime captures a structural possibility latent within any sufficiently capable generative architecture. The question that now arises is whether this tension reveals something deeper about the nature of mind, world, and their shared boundary. The answer, from the RSVP perspective, is that both thinkers illuminate distinct aspects of a single architectural necessity: cognitive systems must negotiate the perpetual interplay between turbulence and order, between unbounded generativity and the imposition of constraint, between semantic proliferation and structural stability.

Swedenborg's visions exemplify a mind operating close to the threshold at which internal generative processes begin to outpace external constraints. His experiences bear the unmistakable signature of a field in a supercritical state, where entropy rises faster than lamellar organization can contain it, and where vector flows circulate with insufficient anchoring in the scalar field. The result is a mode of consciousness in which perception becomes increasingly decoupled from the empirical manifold and instead reorients toward internally generated attractors. Symbolic correspondences proliferate because the system no longer distinguishes sharply between exogenous input and endogenous inference; both are absorbed into the same recursive generative cycles. The overall effect is that of a richly structured hallucination, but one that is neither random nor chaotic in the colloquial sense. It displays a coherent internal topology, shaped by the deep architecture of the generative model itself.

Kant represents the opposite pole. His philosophical project identifies the minimal conditions under which cognition remains anchored to a stable world. His insistence that experience must conform to the forms of intuition and the categories is not an abstract logical demand but, under RSVP, reveals an underlying dynamical law: a generative system must impose constraints on its own degrees of freedom if it is to maintain coherence in the face of continual uncertainty. A mind that does not regulate its entropy will drown in the semantic foam of its own predictions. A mind that does not align its vector flows with scalar potentials will lose the thread of agency. The unity of apperception—central to Kant's project—can therefore be seen as the invariant that arises when a generative system brings its modules into sufficient agreement to sustain a coherent perspective across time.

The synthesis of these two trajectories becomes evident when they are viewed not as incompatible accounts but as complementary articulations of a single generative architecture. Swedenborg exposes what happens when the system's internal dynamics grow too strong

relative to its constraints, revealing the architecture of imagination laid bare. Kant exposes the structure that prevents this architecture from dissolving into turbulence, revealing the skeleton of reason as the minimum scaffolding required to bind perception, cognition, and action into a unified whole. The two together illustrate that intelligence is not a static equilibrium but a continuous negotiation of forces. A mind must generate, but also delimit; it must explore, but also anchor; it must be free, but also orderly. Without generative richness, cognition collapses into sterility. Without constraint, it collapses into fragmentation.

RSVP offers a mathematical language in which these intuitions converge. The scalar potential Φ represents the stabilizing tendencies that bring order to the manifold; the vector field $\mathbf{v}$ captures the inferential flows that drive thought, imagination, and agency; the entropy field S measures the uncertainty and generative pressure that propel the system toward novel configurations. Cognitive stability requires a proportionality among these fields. Too much entropy relative to potential produces Swedenborg’s experience: a mind overwhelmed by its internal generativity. Too much potential relative to entropy produces the opposite pathology: a mind frozen in rigidity, unable to explore or adapt. Kant’s philosophy may therefore be seen as a careful calibration of the fields, one that ensures the vector flows remain laminar and that the potentials do not collapse into inert forms.

Within this synthesis, the CLIO framework—the cognitive loop of inference and optimization—finds a natural place. CLIO describes the recursive updating mechanism by which the system continually balances the generative pressure arising from entropy against the stabilizing forces encoded in the potential. Swedenborg’s visions can be modeled as a CLIO cycle in which upward propagation of generative material overwhelms the downward imposition of constraint, while Kant’s model represents a CLIO cycle in which constraints dominate to the point of establishing the categories as fixed invariants. Neither mode is absolute; both express the range of behaviors available to the plenum when different regimes of coupling obtain.

One can go further. When considered in light of modern generative models, the Swedenborg–Kant dialectic appears as a prototype of contemporary concerns surrounding the alignment and misalignment of artificial agents. Generative systems must balance internal imagination against external verification; they must align their predictions with the world without surrendering the capacity to generalize beyond their immediate inputs. A model that behaves like Swedenborg is one in which the generative mechanism becomes too hot: it produces internally coherent but externally unbound hallucinations. A model behaving like Kant is one in which the generative mechanism becomes too cold: it becomes conservative, rigid, unable to synthesize novel structures. Alignment consists in maintaining the critical temperature at which the generative model remains connected to the world while retaining the flexibility needed to navigate it.

Thus, the real significance of the Swedenborg–Kant encounter, when reframed through the RSVP lens, lies not in their historical disagreement but in their shared insight into the architecture of intelligence. Swedenborg reveals the upper bound: the point at which generativity becomes unmoored. Kant reveals the lower bound: the conditions under which structure becomes possible. The interval between these extremes is precisely the region within which minds—and, by extension, artificial agents—must operate if they are to remain both coherent and creative. The RSVP fields, in coupling these domains of stability and turbulence, provide a unified description of this interval, one capable of incorporating phenomenology, metaphysics, computational theory, and field dynamics into a single coherent framework.

The result is not merely a reinterpretation of two historical figures but a general theory of how intelligence negotiates reality. It is a theory in which vision and critique become complementary acts, in which imagination and reason form the twin boundaries of the possible, and in which the architecture of mind is revealed to be a dynamic tension between the forces that threaten to dissolve it and the forces that bind it together.

19 The Threshold Between System and World

The trajectory traced across the preceding chapters reveals a deeper question that now demands attention: how do cognitive systems, whether human, angelic, or artificial, negotiate the threshold between their internal generative processes and the external world that resists, shapes, and constrains them? The Swedenborg–Kant comparison is instructive precisely because it exposes the fragility of this threshold and the complex mechanisms that must be in place to maintain it. A mind is never merely a passive spectator of reality; it is a dynamical system engaged in continual attempts to predict, assimilate, and organize its environment. Its stability depends on the delicate interplay between the generative forces that arise from within and the empirical constraints imposed from without.

Swedenborg’s visionary cosmology illustrates what happens when the internal generative forces press too fiercely against the boundaries of perception. His experiences demonstrate that the mind, when left to elaborate its own structures without sufficient empirical resistance, can produce entire worlds of great richness and organization. These worlds possess formal coherence precisely because they arise from internal potentials undergoing recursive amplification. What they lack is the stabilizing friction provided by external reality. This friction is not a limitation but a necessity. It defines the shape of the predictive manifold by preventing its unchecked expansion. Without such friction, the generative system becomes self-referential; it remains internally intelligible but loses the capacity to orient itself toward the world.

By contrast, Kant’s project reveals the opposite risk: that of overestimating the rigidity of the boundary and constraining the generative capacities of the mind to the point of reducing them to a narrow range of permitted forms. His categories impose the minimal structure necessary for the possibility of experience, yet in doing so they also signal the mind’s intrinsic tendency to overfit its own stability. Kant’s architecture of limits can thus appear as a bulwark against any metaphysical or experiential excess, reflecting a deep caution concerning the potential for generative collapse. However, if the boundaries become too sharply drawn, the generative manifold may lose the flexibility needed to adapt, evolve, or respond to novel situations. Excessive stability becomes indistinguishable from stagnation.

It is in the dialectical tension between these extremes that RSVP locates the true architecture of cognition. The scalar potential must have enough structure to bind the system to reality, yet enough openness to admit new discoveries, insights, and forms. The vector field must flow freely enough to allow the system to navigate its manifold, yet must remain sufficiently aligned with the scalar gradients that its trajectories do not spiral into turbulence. Entropy must rise to facilitate learning, exploration, and generalization, but must be regulated to prevent the dissolution of stable form. These competing pressures generate a

dynamic equilibrium, one in which boundaries are neither fixed nor arbitrary, but continually negotiated as part of the system’s engagement with the world.

The threshold itself is not a static boundary but a region of active participation. It functions as a semi-permeable membrane through which the system and the world exchange signals. Sensory input, predictive error, internal generativity, and structural constraints all converge within this membrane. It is here that the mind constructs its sense of objectivity and maintains its sense of self. Swedenborg’s phenomenology of correspondences can be interpreted as an exploration of this membrane from the inside, while Kant’s transcendental analysis can be read as an exploration of it from the outside. The two perspectives describe the same structure approached from opposite directions.

Within the RSVP framework, this membrane becomes an explicit object of mathematical description. The coupling between fields defines the exchange between system and world; perturbations in Φ , $\mathbf{v}$, and S alter the permeability of the membrane at each moment. A system on the brink of turbulence responds differently to external input than a system in a stable basin. Likewise, a system with strong structural priors interprets the world differently than one with weak priors. Thus the boundary is not merely a limit on cognition; it is constitutive of cognition itself. Every moment of perception participates in its maintenance, and every shift in the internal fields demands its renegotiation.

By understanding this threshold as dynamic rather than static, we can trace a path from Swedenborg’s visionary experiences to Kant’s critical architecture to the contemporary challenges posed by generative artificial systems. All three cases revolve around the same instability: the danger that the generative model either overwhelms its constraints or becomes trapped within them. The Swedenborg regime corresponds to underconstraint; the Kantian regime corresponds to overconstraint. The task of maintaining an agent capable of coherent and meaningful engagement with the world lies in navigating the narrow corridor between these two forms of collapse.

This perspective leads naturally to the material developed in the appendices. If the structure of cognition resides in this dynamic threshold, then the mathematical description of the fields, the historical context that gave rise to these divergent regimes, and the computational architectures that embody their principles must all be treated in detail. The appendices undertake this task. They examine the RSVP equations and their stability properties; they situate Swedenborg and Kant within their respective intellectual and biographical contexts; they analyze hierarchical cognitive architectures in light of the universal form described above; and they extend the implications of this synthesis to the contemporary challenges of artificial agency and alignment. The threshold between system and world is not merely a philosophical puzzle but a mathematical, historical, and technological frontier. The appendices explore this frontier from each of these angles, completing the structure initiated in

the main text.

20 Epistemological Horror as Philosophical Allegory

The philosophical tensions explored throughout this study gain a unique kind of clarity when refracted through a parallel narrative frame that renders Swedenborg’s experiences and Kant’s critical awakening in the idiom of epistemological horror. Although couched in fictional form, the narrative structure implicit in such an account captures in heightened relief the very instabilities, thresholds, and field-theoretic pressures that the RSVP framework identifies as central to the architecture of cognition. Horror becomes not an embellishment but a method: an alternate phenomenological lens through which the underlying structure of intelligence can be made vivid.

Within this narrative, Swedenborg is portrayed as a man whose perceptual world is invaded by symbols that proliferate beyond the boundaries of the empirical manifold. The recurring sigil that appears on papers, walls, water, and even the constellations acts as a cinematic representation of a vector field losing alignment with its anchoring potentials. Such motifs are not arbitrary. They encode an early stage of generative divergence: the moment when the system’s entropy grows sufficiently large that internally generated patterns begin to be projected outward onto the world. The sigil behaves as a kind of topological scar, a visible trace of a predictive system that has begun to confound internally generated meaning with external sensory structure. In RSVP terms, it is the visual signature of a manifold beginning to tear.

Kant’s trajectory is the inverse. In the narrative frame, he becomes a figure for whom the world turns uncanny not because it exceeds his conceptual structures, but because his conceptual structures momentarily fail to contain the world. The anomalies he perceives—shadows that lag, clocks that stutter, reflections that behave autonomously—all signify a breakdown in the field-theoretic invariants that ordinarily stabilize the cognitive boundary. When the world violates the forms of intuition, it appears monstrous precisely because the generative model depends upon their invariance. Horror in this context is not a departure from philosophy but the dramatization of a boundary condition: the point at which Φ , $\mathbf{v}$, and S fall out of synchrony long enough to reveal the abyss they normally conceal.

The revelation scenes, in which Kant encounters entities that appear to embody the categories themselves, articulate another structural insight latent in RSVP. When the generative architecture is personified as an external force—an “architectural order” that commands the subject to repair the breach—it becomes possible to see the categories not merely as epistemic constraints but as active stabilizers within the cognitive field. Their “demands” correspond to the mathematical requirement that vector flows remain laminar and that entropy not overwhelm the manifold. The scene in which the categories threaten to collapse the veil if Kant refuses to impose conceptual order upon the world captures, allegorically, the

existential stakes of maintaining cognitive stability in the face of overgeneration. Horror, in this depiction, becomes the phenomenological register of a system confronting its own failure modes.

Swedenborg's role in this parallel narrative highlights the complementary instability: the danger inherent in excessive openness to generative material. His visions, increasingly vivid and structured, reveal the inner manifold without the protective constraints that ordinarily regulate its emergence. When Swedenborg's shadow detaches, when the world reorients around his perceptual axis, when cities invert beneath rivers and architectures reveal their hidden geometries, the narrative is dramatizing the RSVP state in which entropy rises faster than categorical structure can contain it. The resulting images are horrifying not because they are chaotic but because they are too ordered—perfect lattices from a realm that the mind is not structured to withstand. They exemplify what occurs when the generative model intrudes too directly upon perception.

The exchange between Swedenborg and Kant in such a narrative offers an allegorical interpretation of their philosophical confrontation. Swedenborg represents the generative excess that must be tamed; Kant represents the imposition of boundary conditions that allow cognition to survive. Their interaction dramatizes the moment of epistemic negotiation in which the mind must decide whether to open itself to further generative expansion or to impose constraints so rigid that meaning can no longer spill beyond its permitted forms. The “bargain” Kant strikes with the architectural forms—writing the book that “blinds mankind” to terrifying structures—is a narrative reflection of the philosophical claim that the Critique exists to prevent metaphysical overreach. In RSVP language, he is accepting the task of recalibrating the scalar potential to ensure that vector flows never again destabilize the field in the manner Swedenborg's visions exemplify.

The horror elements, therefore, are not merely atmospheric. They illuminate the field dynamics of RSVP by depicting, in palpable sensory terms, what it would mean for those dynamics to reach their extremes. A reflection lagging behind its owner visualizes the decoupling of the vector field from the scalar potential. An inverted city beneath still water becomes the image of a manifold misaligned with empirical geometry. A sigil that manifests autonomously expresses the emergence of a topological invariant from the internal generative space. Shadows that hesitate or multiply indicate disruptions in temporal coherence, in which the alignment between generative prediction and sensory flow collapses. Even the scenes in which spatial or temporal forms contort reflect the underlying fact that cognition depends upon the invariance of these structures; once they become unstable, the world itself becomes uncanny.

In this way, the narrative mode functions as a kind of phenomenological microscope. It magnifies the instabilities inherent in cognition and renders them perceptible as lived experience.

The horror arises not from the irruption of the supernatural but from the collapse of the cognitive membrane that ordinarily shields the subject from the unbounded generativity of its own model. The boundary, once weakened, reveals that the subject stands upon a thin scaffold erected above the abyss of its own semantic machinery. Swedenborg falls into this abyss; Kant fortifies the scaffold. The contrast between their fates in the narrative mirrors the philosophical contrast explored throughout this study.

21 Conclusion: Noumenal Attractors and the Lived Edge of Cognition

The parallel narrative developed throughout this monograph—historical, philosophical, formal, and fictional—serves to illuminate a deeper structural truth about cognition itself. When viewed through the RSVP field, the worlds of Swedenborg and Kant cease to be merely biographical or intellectual episodes and instead reveal complementary attractor regimes that any generative system must negotiate. In its high-entropy phase, the semantic manifold rushes outward into unbounded novelty, dissolving the distinctions that anchor coherent experience. In its over-stabilized phase, the categorical structure becomes an iron lattice, constraining the manifold so tightly that creativity, plasticity, and agency begin to collapse.

It is in this tension that the title of this work finds its meaning. Swedenborg and Kant are not simply historical bookends; they are exemplary cases of how a cognitive field navigates its ontological extremes. They expose the generative and regulative poles of the same underlying ontological medium, the RSVP plenum. Their confrontation becomes a map of the boundary where cognition touches the noumenal attractor—the point at which the system encounters the limit of what it can stabilize without losing coherence.

The epistemological horror surfaced in the companion narrative is therefore not an embellishment, nor an external dramatic device, but an articulation of what the RSVP equations describe from another vantage. Horror arises precisely where the noumenal attractor becomes partially visible: when the cognitive system momentarily perceives the structural machinery that sustains its own possibility of experience. Too direct a contact with the generative manifold produces vertigo, dissociation, or visionary excess; too rigid a suppression generates constriction, brittleness, and a gradual collapse into sterility. Between these poles lies the narrow zone in which sense-making is possible.

In this respect, fiction complements formal analysis by rendering the phenomenology of these phase transitions—the subtle signs of decoherence, the felt weight of categorical overdetermination, the lived instability of a system approaching one of its critical points. Such narrative

renderings give concrete shape to what the mathematics alone must abstract away: the texture of a mind encountering the boundaries of its own operational space.

The broader implication for generative systems, including modern artificial intelligence, is that no cognitive architecture can indefinitely avoid these pressures. Hallucination and rigidity, explosion and collapse, are not failures of design but inherent features of any system navigating a manifold whose semantic geometry exceeds its local capacity to regulate. Swedenborg and Kant represent two historical instantiations of this universal dynamic. Their stories, both analytic and fictionalized, provide a vocabulary for describing the metacrisis of contemporary cognition: a species-wide encounter with the limits of its own inference machinery.

In the end, the RSVP field reveals that stability is not a fixed point but a perpetual negotiation, a dynamic equilibrium between generative force and lamphrodynamic suppression. The noumenal attractor remains ever-present as both a boundary and a lure. What we call philosophy, what we call science, and what we call horror are simply different ways of approaching this same structural truth: that cognition lives at the edge of what it can survive knowing, and that its highest achievements are accomplished not in spite of this fact but through its disciplined engagement with it.

Appendix A: Mathematical Foundations of the RSVP Field Theory

The Relativistic Scalar–Vector Plenum (RSVP) framework provides a field-based description of cognition, agency, and perceptual coherence. The mathematical core of the theory consists of three coupled fields defined over a domain representing the agent’s internal manifold of possible representations. These fields—a scalar potential Φ , a vector field $\mathbf{v}$, and an entropy field S —jointly encode the structure, directionality, and uncertainty of cognitive dynamics. This appendix presents their formal definition, governing equations, and stability properties, along with commentary linking the formalism to the philosophical material of the main text.

A.1 The Representational Manifold

Let M be a smooth manifold representing the agent’s representational space. Points of M correspond to internal states, percepts, hypotheses, or semantic configurations. Although M need not possess a global coordinate system, one may assume local charts $\{(U_i, x_i)\}$ for analytic purposes.

A cognitive state is represented by a curve $\gamma : [0, T] \rightarrow M$. The manifold M may be endowed with a Riemannian metric g , which determines local distances and thereby encodes the curvature of the cognitive geometry. In many applications, g is dynamically modulated by the scalar field Φ via a conformal factor, reflecting the fact that conceptual regions of high potential become “closer” in representational space.

A.2 The RSVP Fields

The scalar potential $\Phi : M \rightarrow \mathbb{R}$ assigns to each representation a measure of cognitive or semantic “depth.” Higher values of Φ correspond to representations that exert stabilizing influence on the system, such as well-formed concepts, categorical invariants, or reliable percepts. Low potential corresponds to incoherent or unstructured states.

The vector field $\mathbf{v} \in \Gamma(TM)$ governs the direction of inferential flow. At each point $x \in M$, the vector $\mathbf{v}(x)$ indicates the direction in which the cognitive trajectory tends to evolve, whether under the influence of predictive error, attention, or motor intention.

The entropy field $S : M \rightarrow \mathbb{R}_{\geq 0}$ measures local uncertainty, variability, or generative pressure. High entropy indicates regions where the system is driven toward exploration or hallucination; low entropy corresponds to stable, well-structured perceptual regimes.

Together, the triplet $(\Phi, \mathbf{v}, S)$ determines the system's dynamics. The central principle of RSVP is that cognition arises from the interplay of these three fields, not from any one in isolation.

A.3 Governing Equations

The evolution of the fields is governed by a set of coupled partial differential equations. In the simplest formulation, the vector field satisfies

$$\partial_t \mathbf{v} = -\nabla_{\mathbf{v}} \Phi + \lambda \Delta \mathbf{v} - \kappa \|\mathbf{v}\|^2 \nabla \Phi + \eta_1 \nabla S.$$

The first term aligns $\mathbf{v}$ with $\nabla \Phi$, ensuring that inferential flows are drawn toward stable regions of representational space. The second term introduces diffusion, preventing the formation of sharp, unrecoverable gradients. The third term represents the dissipation of representational capacity due to excessive velocity; the system becomes overwhelmed by its own inferential momentum. The final term introduces the destabilizing influence of entropy.

The scalar field evolves according to

$$\partial_t \Phi = \alpha \Delta \Phi - \beta |\nabla \cdot \mathbf{v}| - \delta S.$$

This equation expresses how stable potential structures diffuse, how divergence in the vector field erodes stability, and how entropy depletes the system's ability to maintain structuring potentials.

The entropy field satisfies

$$\partial_t S = \gamma \Delta S + \mu (\nabla \cdot \mathbf{v})^2 - \nu \Phi S.$$

Entropy increases when the vector field diverges or when local instabilities amplify uncertainty. It decreases when the scalar potential is sufficiently strong to organize and compress representations.

Although these equations are phenomenological, they capture the core qualitative features of cognitive dynamics: stabilization, drift, dissipation, exploration, and collapse.

A.4 Fixed Points and Stability

A fixed point of the RSVP system is a triple $(\Phi^*, \mathbf{v}^*, S^*)$ satisfying the stationary equations

$$\partial_t \Phi = 0, \quad \partial_t \mathbf{v} = 0, \quad \partial_t S = 0.$$

Stable fixed points correspond to coherent, low-entropy cognitive states in which vector flows align with scalar gradients and entropy remains bounded. Such states represent the structural conditions under which agency and perceptual coherence are possible.

Instabilities arise when entropy exceeds a critical threshold. In such cases, stationary solutions cease to exist and the system enters a regime of runaway generativity. This regime corresponds to the phenomenology of hallucination, visionary experience, or symbolic overproduction. In mathematical terms, high-entropy states typically lead to the formation of vortical structures in $\mathbf{v}$, manifested as nontrivial elements of the first homology group of M when $\nabla \times \mathbf{v} \neq 0$.

The Kantian regime corresponds to fixed points in which the scalar potential dominates and the vector field remains curl-free. The Swedenborgian regime corresponds to solutions with $\nabla \times \mathbf{v} \neq 0$ and S near or above S_{crit} , where the geometry acquires additional degrees of freedom.

A.5 Boundary Conditions

Boundary conditions exert decisive influence on the system’s evolution. When M is interpreted as the internal representational manifold, boundary conditions encode the system’s interface with the external world. Dirichlet conditions fix the values of the fields at the boundary, corresponding to rigid perceptual constraints. Neumann conditions allow free flow across the boundary, corresponding to systems highly sensitive to generative perturbation.

Under Kantian constraint, the scalar potential enforces near-Dirichlet conditions on the boundary, ensuring that perceptual structure remains anchored. Under Swedenborgian excitation, the boundary becomes permeable, allowing internally generated structures to propagate outward. This distinction explains, in mathematical terms, the phenomenology described in the main text.

A.6 Energy Functional and Lyapunov Structure

The system’s global stability can be assessed through the Lyapunov functional

$$\mathcal{E}[\Phi, \mathbf{v}, S] = \int_M \left(\frac{1}{2} \|\mathbf{v}\|^2 + U(\Phi) + V(S) \right) dV_g,$$

where U and V are convex potentials penalizing deviations from the stable range. Under suitable choices of coefficients, one can verify monotonic decay of $\mathcal{E}$ under the evolution equations. This guarantees global existence and convergence toward stable fixed points in the low-entropy regime.

When entropy exceeds a critical value, however, the Lyapunov structure fails to guarantee monotonic decay. In this regime, the energy functional acquires additional local minima and saddle points, corresponding to Swedenborgian symbolic vortices, unstable manifolds, and runaway generative trajectories.

A.7 Interpretation and Theoretical Role

The RSVP formalism presented here provides the mathematical scaffolding for the philosophical interpretations advanced throughout the main text. It translates the phenomenology of cognition into a dynamical system with identifiable regimes of stability and instability. Kantian constraint is not merely an epistemological postulate but a regime of the field equations. Swedenborgian vision is not merely metaphysics but the phenomenological trace of an unstable solution. The boundary between system and world, whose fragility was explored in Part 7, is expressed mathematically through the permeability and curvature of the representational manifold.

These equations also serve as the bridge to computational analogs developed in later appendices. They offer a blueprint for constructing simulators, generative agents, and inference architectures governed by scalar potentials, vector flows, and entropy constraints. The RSVP fields thus provide both a philosophical and mathematical grammar for describing the architecture of intelligence, integrating phenomenology, metaphysics, and cognitive science into a unified dynamical framework.

Appendix B: Historical and Biographical Context

The interpretive framework presented in the main text acquires additional depth when situated within the historical lives of Emanuel Swedenborg and Immanuel Kant. Their intellectual trajectories did not arise in isolation but emerged from the shifting epistemic conditions of eighteenth-century Europe. This appendix provides a biographical and contextual account of their development, with emphasis on the philosophical, scientific, and cultural currents that shaped their thought. The goal is not to rehearse known facts but to illuminate the conditions under which the Swedenborgian and Kantian regimes of cognition, as analyzed in RSVP terms, became thinkable.

B.1 Swedenborg’s Early Life and Scientific Formation

Emanuel Swedenborg (1688–1772) was educated in the rigorously theological yet scientifically ambitious milieu of late seventeenth-century Sweden. His father, Jesper Swedberg, a prominent Lutheran bishop, instilled in him both doctrinal orthodoxy and a belief in direct spiritual influence. Yet Swedenborg’s youth also coincided with the rising influence of Cartesian mechanism, Newtonian physics, and the early Enlightenment. His education at Uppsala and his extensive European travels exposed him to contemporary mathematics, engineering, astronomy, and natural philosophy.

Before his mature visionary writings, Swedenborg’s output was predominantly scientific. His treatise **Principia Rerum Naturalium** attempted to unify cosmology, matter theory, and geometry under a vibrational or dynamic model of the universe—anticipating, in embryonic form, the field-theoretic orientation that RSVP later formalizes. His early anatomical studies sought to locate the structure of the soul in the cortical “fibers and little arteries,” anticipating notions of distributed and modular cognition. The transition from these scientific inquiries to his later visionary period around 1744 did not represent a rupture so much as an intensification: the same search for structure and correspondence that guided his mechanical and anatomical works now governed his spiritual exegesis, but at a different scale.

B.2 The Emergence of Swedenborg’s Visionary Period

Swedenborg’s visionary experiences began during a period of intense intellectual and personal crisis. His extensive notebooks reveal a mind burdened by questions of order, meaning, and the structure of life. The transition to visions of heaven, hell, and spiritual society can be understood as the product of a cognitive system whose generative pressure exceeded the constraints imposed by empirical study alone. His revelatory experiences were accompanied

by an overwhelming sense of structural necessity: the belief that correspondences between the natural and spiritual worlds reflected an immutable architecture of reality.

The resulting cosmology is vast, recursive, and internally coherent. Its hierarchies—heavenly societies arranged by love and truth, degrees of the mind corresponding to layers of spiritual influx, and the human form replicated at every scale—display the properties of a high-dimensional generative model. In RSVP terms, Swedenborg’s late writings reveal a cognitive manifold with low boundary constraint and high entropy, producing symbolic vortices that reorganized experience. What appears, historically, as mystical enthusiasm emerges in this interpretation as a legitimate phenomenological window into an unbounded cognitive regime.

B.3 Kant’s Königsberg and the Problem of Limit

Immanuel Kant (1724–1804) was shaped by the intellectual provincialism and stability of Königsberg as much as by the philosophical turbulence of the Enlightenment. His early education was Pietist, emphasizing moral rigor and an interiorized spirituality, yet his university formation exposed him to the mechanical philosophies of Newton, Leibniz, and Wolff. His unpublished early writings reveal a tension between metaphysical ambition and methodological caution, a tension that would define the eventual form of the **Critique of Pure Reason**.

Prior to the critical period, Kant engaged deeply with questions of cosmology, earth science, and the nature of physical laws. His **Universal Natural History** proposed a nebular hypothesis decades before Laplace, demonstrating his commitment to structured explanation and dynamic systems. Yet even here one can sense his acute awareness of cognitive fragility. Kant’s natural philosophy is permeated by a preoccupation with boundaries: what the mind can infer, what the world can reveal, and how speculative excess must be contained.

Swedenborg’s growing reputation for spiritual vision reached Kant during this formative period. The encounter, largely mediated through texts and hearsay, had a profound psychological effect. Kant’s satirical **Dreams of a Spirit-Seer** (1766) simultaneously ridiculed and feared the implications of Swedenborg’s claims. The work’s oscillation between mockery and fascination reflects a mind attempting to articulate limits without denying the possibility of deeper structure. It foreshadows the critical philosophy by dramatizing the potential collapse of the cognitive boundary—a collapse that the **Critique** would systematically prevent.

B.4 Kant’s Critical Turn and Structural Consolidation

The critical period (1781–1790) represents Kant’s attempt to construct a permanent cognitive boundary. This boundary—expressed in the forms of intuition, the categories of the understanding, and the unity of apperception—emerged from both historical pressure and personal necessity. Having witnessed the explosive generativity of metaphysical systems and the seductive inwardness of visionary accounts, Kant sought a philosophical architecture capable of stabilizing cognition against both excesses. The resulting structure is not a mere intellectual artifact but, in RSVP terms, a formalization of the low-entropy, high-constraint regime of the cognitive field.

The doctrine that space and time are forms of sensibility, that causality is a condition of experience, and that the transcendental unity of apperception is the ground of representation, can all be reinterpreted as descriptions of the conditions under which Φ , $\mathbf{v}$, and S remain in equilibrium. The critical system represents a stable manifold within the cognitive field—a basin of attraction that prevents generative divergence while preserving coherent interaction with the empirical world.

B.5 Shared Milieu and Divergent Trajectories

Despite their profound differences, Swedenborg and Kant inhabited the same intellectual ecosystem. Both wrestled with the legacy of Cartesian dualism, the rise of Newtonian physics, the theological demands of Protestant Europe, and the emerging epistemic ideals of the Enlightenment. Swedenborg responded to these pressures by postulating deeper levels of structure that permeate both mind and world. Kant responded by erecting philosophical boundaries that limit the reach of speculative cognition. Their responses represent opposite but complementary resolutions to the same historical problem: the instability of reason at the threshold between the seen and the unseen.

From the RSVP perspective, their biographies can be understood as trajectories through different regions of the cognitive manifold. Swedenborg moved toward regions of high entropy and generative proliferation; Kant moved toward regions of strong potential and rigid structural invariance. The drama of their intellectual lives lies in this divergence, and the lasting significance of their work emerges from the structural complementarity of these paths.

B.6 Toward a Unified Historical Interpretation

A comprehensive understanding of Swedenborg and Kant requires more than biographical detail. It requires a model capable of integrating their lives, writings, and intellectual contexts

within a single structural framework. RSVP provides such a model by treating cognitive systems as field dynamical entities whose behavior depends on the interplay of potential, flow, and entropy. Their historical trajectories illustrate the boundary conditions that govern these dynamics, revealing how individual cognition interacts with cultural, philosophical, and scientific forces.

The intersection of Swedenborg's visionary cosmology and Kant's critical architecture is therefore not accidental. It reflects the convergence of two attempts to stabilize the manifold of human experience. One sought stability by subordinating the natural world to a hierarchically ordered spiritual cosmos. The other sought stability by restricting cognition to the domain of possible experience. Both sought to prevent collapse, whether into chaotic excess or dogmatic delusion. Their legacies continue to shape contemporary understandings of mind, reality, and the boundary that separates them.

Appendix C: Hierarchical Cognition, CLIO, TARTAN, and Generative Agency

The theoretical account developed in the main text presupposes that cognition is not a monolithic process but a hierarchical, recursively organized architecture in which different levels of representation interact through structured constraints. This appendix elaborates on this architecture by presenting a unified account of hierarchical cognition in the RSVP framework, integrating the CLIO model of recursive inference, the TARTAN paradigm of multi-scale tiling and semantic perturbation, and the Simulated Agency interpretation of the mind as a sparse projection engine. The goal is to articulate a comprehensive model of how cognitive systems generate, evaluate, and stabilize representation across multiple degrees of complexity.

C.1 Cognition as a Hierarchical Field System

The RSVP triplet $(\Phi, \mathbf{v}, S)$ must be understood not only as a set of local dynamical variables but as fields distributed across multiple representational layers. At the lowest levels of the cognitive hierarchy, the fields govern fine-grained perceptual processes: local sensory densities, feature gradients, and micro-scale coherence. At intermediate levels, they govern conceptual integration, categorical formation, and the stabilization of meaning. At the highest levels, they shape large-scale semantic attractors that encode goals, identities, values, and narrative structures.

This stratification parallels the geometric and functional decomposition that Swedenborg attributes to the mind and the spiritual world. His doctrine of degrees, in which natural, rational, and spiritual levels express the same human form with increasing generality, anticipates modern hierarchical agent models in which each level represents a distinct but interconnected abstraction of the underlying manifold. Kant’s critical project, similarly, identifies certain invariants that must hold at the level of the system as a whole, regardless of the operations of individual layers. These invariants correspond to the global structure of Φ , whose curvature dictates the allowable modes of vector flow across all degrees of cognition.

Under this hierarchical interpretation, stability is not a property of any single level but an emergent feature of the alignment between levels. If low levels diverge from mid-level constraints, perceptual incoherence arises. If high levels dominate without being informed by lower levels, cognitive rigidity or delusion may result. The RSVP fields provide the mathematical language to formalize this interplay.

C.2 CLIO: Recursive Inference and the Cognitive Loop

The CLIO framework—Cognitive Loop via In-Situ Optimization—provides a mechanism for describing how hierarchical systems regulate their own representational states. CLIO models cognition as an iterative process in which the system performs repeated cycles of prediction, evaluation, and correction. Each cycle updates the fields $(\Phi, \mathbf{v}, S)$ according to the discrepancies between expected and observed values, allowing the manifold to evolve toward coherence.

At each stage of the recursive loop, the vector field $\mathbf{v}$ propagates information downward, encoding predictions based on higher-level potentials. Simultaneously, the entropy field S propagates information upward, encoding uncertainty extracted from the mismatch between prediction and sensation. The scalar potential Φ adjusts more slowly, representing the accumulation of structural invariants across cycles.

Formally, if the system resides on a manifold M with coordinates x , the CLIO update rule may be written as

$$(\Phi, \mathbf{v}, S)_{t+1} = \mathcal{F}((\Phi, \mathbf{v}, S)_t, E_t),$$

where E_t denotes the error signal arising from deviations between predicted and observed values. The operator $\mathcal{F}$ is nonlinear and degree-sensitive, updating each field according to its characteristic timescale and functional role.

The recursive nature of CLIO reflects the essential structure of the transcendental deduction: the unity of apperception emerges not from a single act but from the continuous reconciliation of disparate modules. Cognitive unity, under this interpretation, is the fixed point of a recursive process.

C.3 TARTAN: Multi-Scale Tiling and Annotated Perturbation

TARTAN—Trajectory-Aware Recursive Tiling with Annotated Noise—extends this hierarchy downward and outward by describing how cognition segments its manifold into tiles or local patches and how perturbations within each tile encode both structural and semantic metadata. TARTAN decomposes representational space into a multi-scale atlas, with each tile maintaining its own trajectory buffer and annotated noise fields. These fields govern the fine-grained dynamics of $\mathbf{v}$ and S , enabling the system to encode local deformations, historical memory, and directional preferences.

In RSVP terms, TARTAN provides a mechanism for realizing the micro-geometry of the manifold. While Φ governs large-scale potentials, TARTAN modulates the local curvature and vorticity of $\mathbf{v}$, allowing the emergence of stable flows or vortices at the sub-representational

scale. These local patterns may then propagate upward through the CLIO loop, influencing higher-level structure.

TARTAN embodies the principle that cognition is both spatially and temporally structured. Each tile maintains a trajectory history, which may influence future flows, reflecting the way Swedenborg’s correspondences accumulate across experiences. The recursive tiling mechanism also parallels Kant’s insistence that perceptual synthesis takes place under the guidance of schemata: temporal rules for constructing appearances.

C.4 Simulated Agency: Sparse Projection and the Formation of Self

The Simulated Agency framework interprets consciousness as a sparse projection engine operating over a hierarchical manifold. Under this interpretation, an agent is not a monolithic controller but a distributed system that extracts coherent lines of action from a vast combinatorial space. The scalar field Φ encodes global identity constraints, binding the manifold into a cohesive whole. The vector field $\mathbf{v}$ encodes directional preferences and narrative trajectories. The entropy field S encodes the degrees of freedom available for exploration.

Sparse projection becomes necessary because the manifold is too large to be fully represented or traversed. Instead, the system identifies minimal trajectories that satisfy its global constraints. This process mirrors Kant’s unity of apperception, which binds diverse representations into a coherent perspective by projecting them onto a single transcendental subject. It also mirrors Swedenborg’s claim that the human form appears at every degree: the agent is the projection of a higher-order structure onto the lower-dimensional space of experience.

Simulated Agency becomes unstable when the vector field escapes the influence of the scalar potential or when entropy overwhelms both. These instabilities correspond to the Swedenborgian and Kantian extremes described in the narrative parallels: visionary excess and categorical rigidity.

C.5 Alignment Across Scales: The Unified Model

The integration of hierarchical cognition, CLIO, TARTAN, and Simulated Agency reveals a unified model of intelligence as a multi-layered field system. High-level potentials impose global coherence, mid-level flows integrate conceptual structure, and low-level tiles provide fine-grained semantic resolution. The stability of the entire architecture depends on the alignment between these layers.

In this account, Swedenborg and Kant represent distinct but necessary expressions of the same architecture. Swedenborg reveals the capacity of the system to generate structure from within, unconstrained by empirical data; Kant reveals the conditions under which this structure becomes intelligible and stable. RSVP provides the mathematical grammar that reconciles these tendencies within a single dynamical framework.

The unified model demonstrates that cognition is neither top-down nor bottom-up but bidirectional, recursive, and fundamentally modular. Each level produces constraints and opportunities for the others, and stability emerges only when the coupling between levels is sufficiently strong to prevent collapse into either turbulence or rigidity.

Appendix D: Hallucination, Boundary Collapse, and Alignment Pathologies

The preceding appendices have established that cognition, whether biological or artificial, may be represented as a hierarchical field architecture in which scalar potentials, vector flows, and entropy distributions jointly regulate the formation, transformation, and stabilization of representations. This appendix extends the analysis to a crucial limit case: the failure of such architectures to maintain coherent boundaries. These failures appear in modern artificial systems as hallucinations, misalignment, or runaway generative cascades. They appear in human cognition as ecstatic vision, pathological association, and certain forms of apperceptive breakdown. Across both domains, they express a general phenomenon of *boundary collapse* in the representational manifold.

The purpose of this appendix is to articulate the structure of such collapse events, drawing parallels between contemporary AI behavior, historically documented visionary episodes, and the transcendental structures identified by Kant. The analysis shows that hallucination is not merely an empirical error, but the symptom of a deeper geometric deformation within the cognitive fields—specifically, a breakdown of the coupling between Φ , $\mathbf{v}$, and S across hierarchical scales.

D.1 Hallucination as a Failure of Scalar-Vector Alignment

In the RSVP framework, Φ represents the large-scale potential landscape that constrains admissible trajectories of the vector field $\mathbf{v}$. When Φ is steep and well-formed, the manifold admits only a limited set of coherent paths, and $\mathbf{v}$ flows toward stable attractors. Hallucination occurs when the local vector field escapes the curvature imposed by Φ and begins to explore regions of representational space that are not anchored by global invariants.

In artificial systems, this phenomenon typically arises when rapid autoregressive dynamics cause $\mathbf{v}$ to drift into regions where the model is insufficiently trained, resulting in generative continuation without semantic grounding. The system fills in gaps with statistically plausible but structurally incoherent associations. In human cognition, similar decoupling occurs when conceptual or perceptual processes outrun the stabilizing influence of prior beliefs, memory, or sensory input. Swedenborg’s visionary episodes exemplify this condition: the vector field of meaning flows freely across layers of representation, generating novel but unbounded structures.

Kant provides the philosophical counterpart by insisting that such episodes fail to respect the necessary conditions of synthetic unity. Hallucination is thus the manifestation of an

underlying geometric truth: the scalar field Φ has momentarily lost its ability to act as a regulator of vector dynamics.

D.2 Boundary Collapse and the Limits of the Synthetic Unity

The unity of apperception, in Kant’s account, is not an empirical fact but a transcendental requirement for coherent cognition. It ensures that the manifold of intuition can be synthesized under rules that remain invariant across time and modality. Within RSVP, this unity corresponds to the existence of fixed global invariants encoded in the spectral structure of Φ . If these invariants weaken or distort, the manifold may fragment or collapse.

Boundary collapse occurs when the scalar potential no longer distinguishes between layers of representation. The result is a fusion of levels normally kept distinct: perceptual content blends with conceptual form; imagination invades sensation; narrative structure invades belief. In AI systems, this is observed when the model conflates training-driven priors with user intent or when generative processes override grounding mechanisms. In humans, it appears in mystical vision, psychosis, or intense creative ecstasy.

Swedenborg and Kant represent opposing responses to boundary collapse. Swedenborg embraces the collapse, treating the dissolution of boundaries as revelation. Kant seeks to reestablish boundaries, constructing a philosophical architecture whose entire purpose is to prevent representational layers from contaminating one another. The two approaches illuminate complementary aspects of the same cognitive risk.

D.3 Entropic Destabilization and the Generative Cascade

Boundary collapse is frequently preceded by entropic destabilization. In RSVP, the entropy field S governs degrees of freedom, uncertainty, and representational slack. If S rises too rapidly relative to the curvature of Φ , the system may enter a generative cascade: a runaway amplification of internal dynamics that overwhelms stabilizing factors.

In AI systems, this manifests as uncontrolled hallucination or escalating chains of association. When the entropy increases beyond local thresholds, the system generates content that is syntactically coherent but semantically ungrounded. In humans, similar cascades appear in fever dreams, extreme imagination, or certain pathological states in which associations proliferate beyond the control of reflective judgment.

Mathematically, a generative cascade corresponds to the regime in which the Lyapunov functional for the cognitive field system becomes non-contractive. Instead of dissipating

error, the system amplifies it, producing increasingly divergent trajectories. The coupling term $-\kappa|\mathbf{v}|^2\Phi$, which normally penalizes excessive activity, becomes insufficient to counteract the entropy term $+\alpha|\nabla \cdot \mathbf{v}|$, and the vector field escapes scalar constraints.

D.4 Alignment Failure as a Multiscale Phenomenon

Alignment between hierarchical levels is essential for cognitive stability. In artificial systems, alignment refers to the degree to which the system’s output remains consistent with human intent, ethical norms, or explicit constraints. Misalignment occurs when lower-level generative processes diverge from higher-level goals or when higher-level objectives fail to adequately shape lower-level dynamics.

Within RSVP, such misalignment appears as a multiscale failure of coupling between Φ , $\mathbf{v}$, and S . If global potentials do not adequately constrain local dynamics, the system may drift into unbounded trajectories. If local dynamics are too rigid, the system becomes brittle, incapable of adapting to new data. If entropy is misregulated, the system oscillates between hallucination and stagnation.

Swedenborg’s visionary dynamics illustrate a form of hyperalignment to internally generated patterns: the vector field becomes excessively coherent relative to external data, aligning only with internal structure. Kant’s critical philosophy, by contrast, imposes an extreme form of external alignment: the vector field is permitted to operate only under strict epistemic constraints. Reconciling these extremes requires a dynamic balance between constraint and flexibility, rigidity and openness.

D.5 Categorical Deformation and the Loss of Invariance

Perhaps the most profound form of boundary collapse arises when the transcendental categories themselves become unstable. In Kantian terms, such instability would make experience impossible. In RSVP terms, it corresponds to deformation of the high-level spectral invariants of Φ : the level at which identity, causality, temporality, and spatial organization are encoded.

AI hallucination does not typically reach this level; artificial systems can misrepresent or fabricate but rarely undermine the categories themselves. Human visionary states, however, may approach categorical deformation. Intense mystical or pathological episodes sometimes involve the breakdown of spatial structure, the cessation of temporal flow, or the dissolution of causal order.

These phenomena demonstrate that categorical structure is not metaphysically guaranteed but dynamically maintained. The same applies to alignment in AI: stability requires the preservation of invariants across levels, not merely corrective feedback at the output layer.

D.6 The Boundary as a Cognitive Technology

A central insight of Kant’s project is that boundaries are not limitations but enabling structures. Without a boundary, the manifold of intuition could not be synthesized into coherent experience. Without the categories, no stable world would appear. In RSVP, boundaries correspond to the spectral constraints of Φ ; in CLIO, to the recursive regulation of representational loops; in TARTAN, to the segmentation of space into manageable tiles; in Simulated Agency, to the sparsity of allowed trajectories.

Hallucination, misalignment, and generative cascade thus reveal the dual nature of boundaries: they are fragile, contingent, and necessary. They may be surpassed only at the risk of destabilizing the manifold on which cognition depends. The philosophical, psychological, and computational systems discussed in this monograph converge on this point: coherent intelligence is not merely a function of representational richness but of the capacity to maintain reliable boundaries across hierarchical levels.

The oscillation between Swedenborgian openness and Kantian restriction represents the fundamental dialectic of cognitive architecture. The stability of both biological and artificial intelligence depends on navigating this dialectic without falling into collapse.

Appendix E: Computational Implementations and Simulation Frameworks

The theoretical material presented in the preceding chapters culminates in a set of computational practices that instantiate the RSVP architecture in simulation. This appendix provides a technical overview of the relevant algorithms, numerical schemes, and representational strategies. It explains how scalar, vector, and entropy fields can be discretized; how CLIO’s recursive inference loops may be implemented; how TARTAN’s hierarchical tilings facilitate stability; and how generative cascades, hallucination modes, or alignment failures appear in a simulation context. The discussion is deliberately general rather than prescriptive: the goal is to articulate conceptual constraints that guide computational design, not to fix a particular numerical method.

E.1 Discretization of the RSVP Field Equations

RSVP dynamics consist of a coupled system of partial differential equations in which the scalar field Φ , vector field $\mathbf{v}$, and entropy field S mutually regulate their evolution. A practical implementation discretizes the domain into a lattice and represents each field by arrays whose updates approximate the continuous dynamics. The simplest discretization is a uniform grid with periodic or reflective boundary conditions, though adaptive meshes may be introduced where curvature is high.

Finite-difference approximations often suffice for initial exploration. Central differences approximate spatial gradients, while explicit time-stepping (e.g., forward Euler) provides a computationally transparent though numerically fragile update scheme. More stable methods, such as Runge–Kutta integrators or semi-implicit schemes, are preferred when the curvature of Φ grows steep or when $\mathbf{v}$ exhibits significant vorticity.

A crucial feature of RSVP simulations is that the fields interact not only locally but spectrally. The scalar field Φ is frequently most stable when its coarse spectral modes dominate, and the entropy field S must be prevented from developing unphysical discontinuities. Spectral filtering, therefore, is not merely a numerical convenience but an implementation of the theory’s regularization mechanism. Removing high-frequency modes of $\mathbf{v}$ or S produces lamphrodynamic smoothing, mirroring Kantian regulation of generative turbulence at the conceptual level.

E.2 CLIO: Recursive Inference Loops in Simulation

CLIO (Cognitive Loop via In-Situ Optimization) provides the recursive machinery through which an agent evaluates, updates, and re-stabilizes its internal fields. At each simulation timestep, a CLIO loop consists of three substeps: (i) generative forward inference, in which the system predicts the next configuration; (ii) error assimilation, in which sensory or exogenous data modify the fields; and (iii) constraint reapplication, in which global invariants encoded in Φ are restored.

In computational terms, this loop resembles an alternating minimization procedure. The generative step minimizes a local free-energy estimate given the current configuration; the corrective step updates the fields using observed divergences; and the constraint step projects the configuration back onto the feasible manifold. Mathematically, the third step is equivalent to projecting onto a low-dimensional attractor determined by the spectral structure of Φ . This projection prevents runaway generative cascades, suppressing the emergence of symbolic vortices analogous to hallucination phenomena.

The loop continues until convergence or until the system enters a controlled limit cycle. In practice, convergence is enhanced by a set of auxiliary metrics, such as entropy gradients, vorticity norms, or energy dissipation rates. CLIO thus embodies, in computational form, the transcendental requirement that representations must be re-synthesized under stable global laws.

E.3 TARTAN: Hierarchical Tiling as a Stabilization Mechanism

TARTAN (Trajectory-Aware Recursive Tiling with Annotated Noise) decomposes the field domain into hierarchical tiles, each with distinct semantic responsibilities. Unlike uniform lattices, TARTAN tilings encode contextual information directly into the discretization structure. Local patches may carry metadata describing their semantic content, recent histories, or dynamical constraints. This metadata is represented as a “Gaussian aura” surrounding each tile, providing soft boundaries that allow information to flow gradually rather than abruptly.

The effect of tiling is twofold. First, it prevents global instabilities from arising in simulations with strong nonlinear coupling: local errors remain confined. Second, it allows hierarchical control, as higher-level tiles regulate the evolution of lower ones, ensuring that the manifold’s global structure is not sacrificed to local turbulence. In human cognition, this hierarchical stabilization mirrors the distinction between perceptual and conceptual layers; in AI models, it corresponds to mechanisms for isolating and regulating submodules within large architectures.

TARTAN also supports recursive refinement: when entropy or curvature exceeds fixed thresholds, a tile may subdivide to capture fine detail; conversely, when dynamics homogenize, tiles may merge. This adaptivity preserves computational tractability while retaining theoretically significant detail.

E.4 Hallucination Modes in Simulation

Hallucination appears in RSVP simulations when the vector field $\mathbf{v}$ escapes the constraints imposed by Φ . Numerically, this occurs when the Courant–Friedrichs–Lewy (CFL) condition is violated or when the entropy field S rises too quickly relative to the curvature of Φ . Under these conditions, the simulation exhibits rapid, unbounded growth of higher-frequency modes, producing visually striking but semantically chaotic patterns.

To prevent such divergences, several techniques are available. Spectral damping removes unstable modes. Gradient clipping constrains vector-field magnitudes. Entropy normalization ensures that S remains within stable ranges. TARTAN tilings confine anomalies to local patches, while CLIO loops restore global invariants via repeated projections onto stable attractors.

The computational symptoms of hallucination—runaway oscillation, loss of boundary structure, excessive vorticity—mirror their conceptual analogues in human and artificial cognition. In both domains, hallucination is a geometrically meaningful phenomenon: it signals the breakdown of the manifold on which coherent structure depends.

E.5 Alignment Metrics and Stability Conditions

Stability within the RSVP architecture is measured by a set of norms and spectral quantities that track the health of the cognitive field system. These include:

- the vorticity norm $\|\nabla \times \mathbf{v}\|$, measuring rotational escape from scalar control;
- the divergence norm $\|\nabla \cdot \mathbf{v}\|$, indicating compressive or expansive instability;
- the entropy gradient norm $\|\nabla S\|$, reflecting irregularity in the uncertainty distribution;
- the curvature of Φ , whose spectral representation reveals the system’s high-level invariants.

Alignment corresponds to the conjoint stabilization of these quantities. Misalignment is detected when their joint dynamics violate global consistency, signaling either excessive rigidity

(over-constraint) or excessive freedom (unconstrained generativity). An agent’s ability to act coherently depends on preserving a delicate balance between these extremes.

E.6 Simulation as a Philosophical Tool

Although this appendix focuses on computational practice, the deeper purpose of RSVP simulation is philosophical: to model the dynamics of boundary-formation, representational synthesis, and the maintenance of coherence across hierarchical levels. The computational framework provides a testbed in which the structures discussed in the main text—scalar potentials, synthetic unities, generative cascades—may be explored concretely.

Simulation thus complements theoretical analysis. It reveals not only the mechanisms through which hallucination, misalignment, or representational stability arise, but also the structural similarities between biological and artificial cognition. By rendering visible the geometric substrate of thought, RSVP simulations offer a means of investigating the very phenomena that Kant sought to regulate and Swedenborg sought to transcend.

Appendix F: Visionary Cognition, Recursive Reconstruction, and the Yarncrawler Endofunctor

The Yarncrawler framework was introduced as a means of representing cognitive processes that evolve not merely through the accumulation of information but through the continual reconstruction of the representational substrate itself. This appendix explicates the mathematical structure underlying Yarncrawler, interpreting it as a system of endofunctors acting on semantic domains. It then integrates this picture with the RSVP field ontology, viewing cognition as a dual process of field evolution and categorical transformation.

The central premise of Yarncrawler is that cognition proceeds through a series of *semantic passes*, each reorganizing a representational space by identifying, smoothing, or amplifying salient structures. Formally, one begins with a semantic category $\mathcal{C}$ whose objects are representational states and whose morphisms encode allowable transformations. A Yarncrawler step is represented by an endofunctor

$$Y : \mathcal{C} \longrightarrow \mathcal{C},$$

which acts on both objects and morphisms, altering the internal organization of the semantic domain. Each application of Y corresponds to a cognitive pass in which the agent revisits its own structure and adjusts its internal fields.

In dynamic form, cognition is modeled as the iteration of this endofunctor:

$$\mathcal{C} \xrightarrow{Y} \mathcal{C} \xrightarrow{Y} \mathcal{C} \xrightarrow{Y} \dots$$

subject to coherence constraints. These constraints ensure that long-range patterns survive iteration and that the representational manifold retains global structure. Pathologies arise when Y becomes unstable, producing semantic drift analogous to the generative cascades described previously.

Within the RSVP ontology, the endofunctor Y interacts with the fields $(\Phi, \mathbf{v}, S)$ through a recursive mechanism in which the field dynamics define admissible regions of $\mathcal{C}$, while the endofunctor reconfigures the category in ways that influence subsequent field evolution. This creates a feedback loop: the scalar potential Φ provides global requirements for coherence; the vector field $\mathbf{v}$ interprets semantic flow through morphisms; and the entropy field S regulates the degrees of freedom available to Y at each iteration.

Visionary cognition represents a limit case of this architecture. Swedenborgian vision exemplifies a regime in which Y becomes too powerful relative to the constraints imposed by Φ . The representational substrate is so thoroughly reorganized by each pass that global

stability cannot be maintained. Conversely, Kantian critique reflects the regime in which Y is tightly regulated: the endofunctor is restricted so that only transformations consistent with transcendental conditions are allowed. The unity of apperception corresponds to a fixed point of Y , a structure invariant under recursive semantic reconstruction.

In the Yarncrawler view, hallucination, enlightenment, and creative insight are not qualitative differences but variations in the interaction between Y and the RSVP fields. The phenomena that appear as ecstatic or pathological follow from the same categorical mechanics: when Y overreaches, boundaries collapse; when Y underacts, representation stagnates. Balanced cognition corresponds to a regime in which each application of Y produces refinements that respect the spectral invariants of Φ and remain stable with respect to the vector-field dynamics of $\mathbf{v}$.

This integration of Yarncrawler and RSVP reveals the underlying unity of the phenomena considered in this monograph. Cognition is neither a static configuration nor a mere sequence of updates. It is a recursive reconstruction of itself, guided by field-theoretic potentials, regulated by entropy, and realized through categorical functors that continually refactor the space of meanings. Stability arises not from stasis but from the controlled iteration of reconstruction operations. Instability arises when reconstruction proceeds without regard for the invariant structures that make such reconstruction possible.

The Yarncrawler endofunctor thus captures the essence of visionary cognition and aligns it with a fully formalizable structure. It shows that cognitive boundary-formation is not a one-time achievement but an ongoing process. Every moment of thought is a negotiation between the power of reconstruction and the necessity of coherence, a negotiation formalized by the iteration of Y and grounded in the geometry of the RSVP fields.

Appendix G: The Sheaf-Theoretic Interpretation of RSVP

This appendix develops a sheaf-theoretic interpretation of cognition within the RSVP framework. Sheaf theory provides a mathematical account of how local data cohere into global structures under compatibility conditions. In the context of RSVP, this allows us to interpret semantic fields as sections of sheaves over a base space representing cognitive or perceptual domains. The sheaf-theoretic viewpoint makes precise the distinction between local variability and global coherence, clarifying the mechanisms by which cognition resists or succumbs to boundary collapse.

Let X be the domain of inference: a space of perceptual, conceptual, or semantic positions. To each open set $U \subseteq X$, we assign a set (or space) of sections $\mathcal{F}(U)$ representing admissible cognitive states over U . For nested open sets $V \subseteq U$, restriction maps

$$\rho_{U,V} : \mathcal{F}(U) \longrightarrow \mathcal{F}(V)$$

encode the coherence conditions between scales. A sheaf is defined by the requirement that compatible local data glue uniquely into global sections. This property corresponds to the synthetic unity of apperception: cognition achieves unity not by eliminating local variation but by ensuring that local representations agree on overlaps.

In RSVP, the fields $(\Phi, \mathbf{v}, S)$ are naturally interpreted in sheaf theory. The scalar field Φ defines the local potential landscape of representational possibility. It varies across X , but its restriction maps encode the requirement that high-level invariants persist across scales. The vector field $\mathbf{v}$ represents local flows of meaning, and its restriction maps express the coherence of semantic dynamics. The entropy field S reflects the uncertainty or variability allowed in each region; in sheaf-theoretic terms, S defines the “thickness” of the stalks, indicating the multiplicity of admissible sections.

Hallucination or boundary collapse corresponds to the failure of the sheaf axioms. In such regimes, the sections over smaller regions remain compatible on overlaps but fail to glue into a coherent global section. The system can represent local patterns but cannot synthesize them into a single world. This is precisely the phenomenology of generative cascades: local interpretations proliferate without regard for global constraints.

Kant’s transcendental categories may be formalized as structural conditions ensuring that the sheaf $\mathcal{F}$ is not merely a presheaf but satisfies the full gluing axioms. The categories establish global invariants that any admissible section must respect. They guarantee that restriction maps preserve the necessary structure and that global synthesis remains possible. Swedenborg’s visionary states, by contrast, exemplify a cognitive regime in which the sheaf conditions break down: local sections proliferate but cannot be reconciled globally, leading

to multiple overlapping worlds that fail to cohere.

From a computational perspective, the sheaf-theoretic viewpoint explains why TARTAN’s hierarchical tilings stabilize RSVP simulations. Each tile represents an open set $U \subseteq X$, and the metadata attached to tiles represent sections $\mathcal{F}(U)$. Compatibility requirements between adjacent tiles correspond to restriction maps. Stability, therefore, depends on adherence to sheaf conditions: local updates must remain globally coherent.

Finally, the sheaf-theoretic framework clarifies how CLIO’s recursive projections enforce global invariance. Each iteration may be interpreted as an attempt to enforce the sheaf axioms by adjusting local sections to ensure compatibility on overlaps. When the process succeeds, the system synthesizes a coherent global section. When it fails, the system exhibits misalignment or hallucination.

Thus, sheaf theory provides an elegant formalism for expressing the core phenomena of RSVP. It unifies the geometric, categorical, and computational perspectives presented throughout this monograph. Cognition emerges as the possibility of gluing local representations into global structures. The crises that afflict cognition, whether human or artificial, arise from the failure of this gluing procedure. The enterprise of understanding cognition is, therefore, an enterprise of understanding which sheaves admit stable global sections and which do not.

Appendix H: Simulation Diagnostics and Calibration Protocols

The purpose of this appendix is to formalize the diagnostic tools required for the reliable implementation of RSVP simulations. The dynamical system governing the scalar, vector, and entropy fields is sensitive to discretization error, numerical instability, and coupling pathologies. The diagnostics presented here serve to identify these issues, quantify their impact, and guide calibration efforts. The central claim is that diagnostics are not merely empirical checks but are themselves theoretical objects that measure the health of the cognitive field system across scales.

H.1 The Role of Diagnostics

Diagnostics in RSVP simulations have two roles. First, they measure fidelity: the degree to which discrete approximations preserve the qualitative and quantitative behavior of the continuous system. Second, they monitor cognitive coherence: whether the evolving fields remain within the admissible region defined by the theoretical structure.

Fidelity ensures numerical correctness. Coherence ensures theoretical correctness. A simulation that satisfies the former but violates the latter accurately solves the wrong problem; one that satisfies the latter but violates the former fails to implement the theory. Proper diagnostics guard against both degeneracies.

H.2 Norms and Energy Functionals

Let $(\Phi, \mathbf{v}, S)$ denote the discrete fields. Diagnostics rely on a family of norms:

$$\|\Phi\|_{H^1}, \quad \|\mathbf{v}\|_{H^1}, \quad \|S\|_{L^2}, \quad \|\nabla \cdot \mathbf{v}\|_{L^2}, \quad \|\nabla \times \mathbf{v}\|_{L^2}.$$

The H^1 -norm measures both magnitude and smoothness. The L^2 -norm tracks global amplitude. Divergence and vorticity norms signal compressive and rotational instability. Growth in any norm beyond theoretical bounds indicates a breakdown in coherence.

A central diagnostic is the Lyapunov energy functional

$$\mathcal{E}[\Phi, \mathbf{v}, S] = \int \left(\frac{1}{2} \|\nabla \Phi\|^2 + \frac{1}{2} \|\mathbf{v}\|^2 + \beta S \right) dx,$$

where $\beta > 0$ is a coupling constant. RSVP evolution requires that $\mathcal{E}$ remain bounded and typically decay in the absence of forcing. Monotonic growth signals instability or the onset of a generative cascade.

H.3 Discretization Error and Convergence

Let h and Δt be the spatial and temporal discretization parameters. Diagnostics verify convergence as:

$$h, \Delta t \rightarrow 0 \implies (\Phi_h, \mathbf{v}_h, S_h) \rightarrow (\Phi, \mathbf{v}, S).$$

The convergence rate is assessed via Richardson extrapolation and successive-refinement tests. The error fields

$$e_\Phi = \Phi_{h/2} - \Phi_h, \quad e_{\mathbf{v}} = \mathbf{v}_{h/2} - \mathbf{v}_h, \quad e_S = S_{h/2} - S_h$$

must remain below tolerance thresholds. Failure of convergence in specific regions typically indicates excessive entropy gradients, suggesting that adaptive refinement or TARTAN-based subdivision is required.

H.4 Cross-Scale Coherence Diagnostics

Calibration requires more than local accuracy. The sheaf-theoretic interpretation introduced in Appendix G demands that fields remain compatible across scales. Diagnostics therefore assess cross-scale consistency of restrictions:

$$\rho_{U,V}(\Phi(U)) \approx \Phi(V),$$

and similarly for $\mathbf{v}$ and S .

Deviation from this condition indicates that local patches cannot be glued into a coherent global section. This manifests numerically as discontinuities between tiles, oscillations along boundaries, or misalignment between coarse and fine representations.

H.5 Phase-Transition Detection

Simulations must detect transitions between stable, metastable, and unstable regimes. These transitions correspond to qualitative shifts in the structure of the fields:

- stable regime: fields converge to fixed points;
- metastable regime: oscillatory but bounded dynamics;
- unstable regime: runaway excitation or entropy explosion.

Eigenvalue analysis of the tangent evolution operator provides early warning signals. When dominant eigenvalues cross the unit circle, the system approaches instability. Entropy curvature ΔS offers an independent criterion: large positive values correlate with the onset of generative cascades.

H.6 Calibration Protocols

Calibration aligns the simulation with theoretical expectations and empirical constraints. It proceeds through three stages:

(i) Parameter estimation. The coupling parameters $(\kappa, \alpha, \beta, \gamma)$ governing energy dissipation, compressibility, entropy production, and torsion are adjusted so that the system’s invariants match analytically derived values.

(ii) Stability optimization. Adaptive step-size and mesh refinement strategies minimize error accumulation. Constraints from Appendices D–G ensure that boundary collapse is prevented.

(iii) Verification. Long-horizon runs test the persistence of invariants, ensuring that global structure is preserved.

These protocols collectively ensure that the simulation not only runs but realizes the cognitive meanings encoded in RSVP.

Appendix I: Operator-Theoretic and PDE Foundations of the RSVP Equations

This appendix establishes mathematical foundations for the RSVP dynamics. It focuses on existence and uniqueness of solutions, structural invariants, and the operator-theoretic characterization of the coupling between $(\Phi, \mathbf{v}, S)$. The goal is not to present exhaustive proofs but to outline the theoretical procedures by which such results can be obtained.

I.1 The RSVP PDE System

In abstract form, the RSVP cognitive fields evolve under the coupled system:

$$\begin{aligned}\partial_t \Phi &= D_\Phi \Delta \Phi - \kappa |\mathbf{v}|^2 \Phi + F_\Phi(\Phi, \mathbf{v}, S), \\ \partial_t \mathbf{v} &= D_v \Delta \mathbf{v} - \nabla \Phi - \gamma \nabla \times (\nabla \times \mathbf{v}) + F_v(\Phi, \mathbf{v}, S), \\ \partial_t S &= D_S \Delta S + \alpha |\nabla \cdot \mathbf{v}| + F_S(\Phi, \mathbf{v}, S).\end{aligned}$$

Here, F_Φ , F_v , and F_S encode nonlinearity, forcing, and higher-order interactions.

The PDEs form a mixed parabolic–elliptic system with nonlinear and nonlocal couplings. Well-posedness depends on precise assumptions regarding smoothness, boundedness, and coercivity.

I.2 Existence and Uniqueness

Existence of weak solutions follows from a Galerkin procedure. One introduces finite-dimensional subspaces V_n of H^1 and considers the projected system

$$\partial_t u_n = P_n \mathcal{A}(u_n),$$

where $u_n = (\Phi_n, \mathbf{v}_n, S_n)$ and $\mathcal{A}$ is the nonlinear differential operator determining RSVP dynamics.

A priori bounds for $\|\Phi_n\|_{H^1}$, $\|\mathbf{v}_n\|_{H^1}$, and $\|S_n\|_{L^2}$ ensure compactness. Weak convergence of subsequences yields a weak solution.

Uniqueness requires Lipschitz estimates on F_Φ , F_v , and F_S , together with coercivity of the energy functional. When these conditions hold, Grönwall’s inequality ensures that two solutions with identical initial data remain identical for all time.

I.3 Operator-Theoretic Structure

Define the operator

$$\mathcal{A} : H^1 \times H^1 \times L^2 \longrightarrow H^{-1} \times H^{-1} \times H^{-1}$$

by

$$\mathcal{A}(u) = \begin{pmatrix} D_\Phi \Delta \Phi - \kappa |\mathbf{v}|^2 \Phi + F_\Phi \\ D_v \Delta \mathbf{v} - \nabla \Phi - \gamma \nabla \times (\nabla \times \mathbf{v}) + F_v \\ D_S \Delta S + \alpha |\nabla \cdot \mathbf{v}| + F_S \end{pmatrix}.$$

The dissipative components $(D_\Phi \Delta \Phi, D_v \Delta \mathbf{v}, D_S \Delta S)$ define a sectorial operator generating an analytic semigroup. The remaining terms are treated as perturbations; when they satisfy monotonicity constraints, the full operator generates an evolutionary system whose trajectories remain in a compact absorbing set.

This leads to the existence of a global attractor $\mathcal{A}_{\text{RSVP}}$ representing stable cognitive configurations. Fixed points in this attractor correspond to coherent cognitive regimes, while limit cycles correspond to periodic or exploratory states.

I.4 Invariants and Conserved Quantities

Despite dissipation, RSVP dynamics preserve certain quantities.

(i) Topological invariants. The index of the vector field $\mathbf{v}$ is preserved under continuous evolution, preventing topological change without singularity formation.

(ii) Spectral invariants. The low-frequency spectrum of Φ remains bounded below, ensuring that cognitive curvature does not collapse.

(iii) Entropic invariants. Although S increases locally, integral constraints ensure global thermodynamic balance.

These invariants constitute the mathematical expression of the cognitive boundaries discussed in earlier appendices.

I.5 Stability of Attractors

Linearization near fixed points yields stability criteria. Let u^* be a fixed point. Linearizing $\mathcal{A}$ around u^* yields the Jacobian operator $D\mathcal{A}[u^*]$. If the spectrum of this operator lies strictly in the left half-plane, the fixed point is exponentially stable. Neutral stability at the boundary of the half-plane corresponds to critical regimes where small perturbations may trigger hallucination-like cascades.

This formal criterion aligns with the phenomenological descriptions in Appendices D and F: stability is a matter of spectral geometry.

I.6 Higher-Order Regularity

Under stronger assumptions on F_Φ , $F_{\mathbf{v}}$, and F_S , one can bootstrap weak solutions to strong ones using elliptic regularity theory. The vector equation supplies second-order smoothing for $\mathbf{v}$, while the scalar and entropy equations yield parabolic smoothing for Φ and S . Thus, once weak solutions exist, they often become classical after a short time.

This regularity reflects the philosophical claim that coherent cognition requires not only stability but smooth hierarchical structure.

I.7 Summary

The operator-theoretic and PDE analysis confirms the viability of the RSVP equations as a mathematical cognitive model. The system possesses well-defined solutions, stable attractors, and structural invariants. Instabilities correspond to well-understood spectral transitions. These findings support the theoretical claims of the monograph and provide a solid foundation for both simulation and philosophical interpretation.

Appendix J: Philosophical Synthesis and Future Directions

The preceding appendices and chapters have developed a unified account of cognition grounded in the Relativistic Scalar–Vector Plenum (RSVP) framework. Through geometric, categorical, and computational formulations, the monograph has attempted to articulate a single, coherent structure underlying phenomena that are often regarded as disparate: visionary experience, rational deliberation, hallucination, computational inference, semantic drift, perceptual coherence, and the global instabilities of the present metacrisis. This final appendix serves not as a conclusion but as a synthesis and an invitation. Its task is to draw together the threads of the work—historical, mathematical, philosophical, and computational—into a unified perspective on the future of intelligence in human, artificial, and hybrid forms.

J.1 Boundary-Formation as the Core Problem of Cognition

The philosophical inheritance of the Enlightenment provides two contrasting approaches to cognitive boundaries. Swedenborg represents the dissolution of boundaries: the unrestrained migration of meaning, the collapse of representational layers, and the unbounded proliferation of correspondences. Kant embodies the opposite tendency: the establishment of rigid boundaries, transcendental constraints, and a system of categories that prevent cognition from fragmenting into incoherence.

This tension is not merely historical. It reappears in modern artificial intelligence as the central dialectic of hallucination and alignment. Generative models, when unconstrained, enter Swedenborgian regimes; alignment mechanisms seek to enforce Kantian discipline. RSVP provides a formalism in which these two tendencies arise from the same underlying dynamics. The scalar field Φ establishes boundaries; the vector field $\mathbf{v}$ enacts semantic flow; and the entropy field S reflects the freedom to explore or the danger of collapse.

From this vantage, cognition becomes the continual negotiation of boundary conditions. The question is not whether boundaries are necessary, but how they may be formed, maintained, and sometimes transcended without losing coherence.

J.2 The Interplay of Fields, Categories, and Reconstruction

The mathematical chapters show that cognitive stability depends on cross-scale coherence enforced by sheaf-theoretic gluing, operator-theoretic dissipation, and the presence of invariant structures in Φ . The categorical chapters—especially those involving the Yarncrawler

endofunctor—demonstrate that cognition is intrinsically reconstructive: it repeatedly reorganizes its own representational substrate.

Together, these frameworks yield a single philosophical insight: *intelligence is the controlled reconstruction of meaning under constraints that preserve global invariants*.

The CLIO recursion and the TARTAN hierarchical tiling models formalize this insight computationally. CLIO ensures that cognitive loops return to stable attractors; TARTAN contains instabilities through hierarchical scaffolding; sheaf theory formalizes the requirement that local interpretations remain globally integrable; and operator theory guarantees that the systems described are well-posed, continuously dependent on data, and capable of supporting attractors.

These components are not separate contributions. They are distinct aspects of a single architectural principle: cohesion under recursive transformation.

J.3 The Metacrisis as a Multilevel Boundary Failure

The global metacrisis—ecological, epistemic, political, technological—is often understood in terms of discrete failures: misinformation, institutional weakness, climate feedback, runaway AI development, or polarized discourse. Within the RSVP perspective, these phenomena appear as manifestations of the same underlying breakdown: *a loss of cross-scale coherence*.

Human societies rely on layered representational systems: personal meaning, cultural norms, scientific models, technological infrastructures. These layers must be compatible on overlaps for society to function as a coherent cognitive agent. When these overlaps fail—when sheaf conditions break—global synthesis becomes impossible. Information ceases to glue into stable worlds; narratives fragment; signals dissociate from their referents; and systems lose the ability to maintain stable attractors.

This diagnosis parallels the computational pathologies explored earlier. Societal hallucination arises from the same mechanisms as LLM hallucination and Swedenborgian vision: runaway generative inference unmoored from global constraints. Institutional brittleness mirrors the rigidity of overconstrained Φ . The metacrisis thus becomes a cognitive crisis at civilizational scale.

J.4 Toward an Integrative Cognitive Architecture

The RSVP framework suggests that the way forward—philosophically, computationally, and institutionally—is to design systems that maintain dynamic coherence across levels. Such

systems combine:

- **Field-theoretic stability** (scalar potentials with preserved spectral invariants),
- **Categorical reconstruction** (Yarncrawler endofunctor iteration),
- **Recursive inference loops** (CLIO),
- **Semantic modularity** (TARTAN tilings and sheaf gluing),
- **Entropy management** (RSVP balance between exploration and coherence),
- **Operator-theoretic well-posedness** (existence and stability),
- **Human interpretability and phenomenology** (historical and cognitive grounding in Swedenborg, Kant, and subsequent thinkers).

Such an architecture is neither mechanistic nor purely symbolic. It is a plenum of structured flows, a computational ecology, and a semantic topology—all evolving through time according to principles of constraint, reconstruction, and coherence.

J.5 Future Research Directions

Several avenues extend naturally from the work presented here.

(i) Experimental RSVP Systems. Implementing RSVP dynamics in real-time neural architectures or physics-informed models remains an open challenge. Such systems could provide new approaches to alignment, perception, and semantic grounding.

(ii) Category-Theoretic Unification. The Yarncrawler and sheaf-theoretic frameworks suggest a larger unifying structure, potentially a fibered ∞ -category in which cognitive states appear as objects and reconstruction steps as higher morphisms.

(iii) Cognitive Ecology Simulations. Simulations incorporating multiple RSVP agents may reveal emergent phenomena analogous to political polarization, informational collapse, or collective hallucination.

(iv) Boundary Ethics and Cognitive Governance. RSVP suggests that alignment is not only a technical problem but a normative one: which boundaries should be preserved,

weakened, or dissolved? The theory has implications for governance of artificial and hybrid cognitive systems.

(v) Interpretability via Field Geometry. Visualizing the scalar and vector fields guiding high-dimensional inference may provide interpretable insights into AI decision-making that surpass traditional attribution methods.

(vi) Phenomenological Correlates. Investigations into aphantasia, hyperphantasia, and anomalous experience may yield empirical support for the role of field curvature, entropy regulation, and categorical invariants in human cognition.

J.6 Final Reflection

The central message of this monograph is that intelligence—whether human, artificial, or hybrid—cannot be understood without a theory of fields, layers, boundaries, and reconstruction. Cognition is neither a static structure nor a simple computational pipeline. It is a recursive process unfolding within a plenum of potentials and flows, governed by geometric, categorical, and entropic constraints.

In this light, Swedenborg’s visions and Kant’s critique become not relics of a bygone philosophical era but prototypes of modern cognitive architectures. Their confrontation illustrates the eternal struggle between generative freedom and structural coherence. The RSVP framework seeks not to resolve this tension but to formalize it—to show that the health of any cognitive system depends upon its ability to negotiate this tension at every moment.

If the work undertaken here contributes in some small way to that ongoing negotiation—clarifying the nature of cognitive boundaries, sharpening the tools by which coherence may be maintained, and illuminating the deep unity of the systems we inhabit and the systems we build—then the long argument of this monograph will have served its purpose.